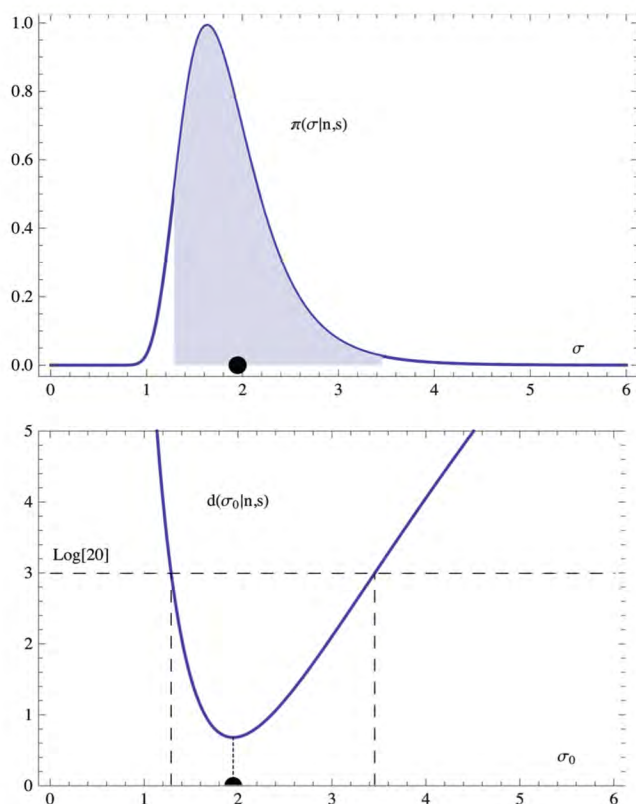


CURRENT TRENDS IN BAYESIAN METHODOLOGY WITH APPLICATIONS



Edited by
Satyanshu Kumar Upadhyay
Umesh Singh
Dipak K. Dey
Appaia Loganathan

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Dedicated to my Parents and my Teacher Prof. Manju Pandey

– **Satyanshu K. Upadhyay**

Dedicated to my late Mother Mrs. Siddheswari Devi

– **Umesh Singh**

Dedicated to my late Parents Mr. Debendranath Dey and Mrs. Renuka Dey

– **Dipak Dey**

Dedicated to my Teachers Prof. M. Rajagopalan and Prof. G. Nanjundan

– **Appaia Loganathan**

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Contents

List of Figures	xix
List of Tables	xxvii
Preface	xxxi
Foreword	xxxiii
List of Contributors	xxxv
1 Bayesian Inference on the Brain: Bayesian Solutions to Selected Problems in Neuroimaging	1
<i>John A. D. Aston and Adam M. Johansen</i>	
1.1 Introduction	2
1.2 Background	3
1.2.1 Neuroscience and its imaging problems	3
1.2.2 Monte Carlo and sequential Monte Carlo	4
1.3 Application: Functional Magnetic Resonance Imaging	8
1.3.1 Problem	9
1.3.2 Methodology	10
1.3.3 Results	11
1.4 Application: Positron Emission Tomography	15
1.4.1 Problem	15
1.4.2 Background	16
1.4.3 Modelling & Methodology	18
1.4.3.1 Modelling	18
1.4.3.2 Methodology	19
1.4.4 Example	21
1.5 Application: Magnetoencephalography	23
1.5.1 Problem	23
1.5.2 Background	23
1.5.3 Modelling & Methodology	24
1.5.3.1 Modelling	24
1.5.3.2 Methodology	26
1.5.4 Example	27
1.6 Conclusions	29

2	Forecasting Indian Macroeconomic Variables Using Medium-Scale VAR Models	37
	<i>Goodness C. Aye, Pami Dua, and Rangan Gupta</i>	
2.1	Introduction	37
2.2	Econometric Models	39
2.2.1	VAR and BVAR	39
2.2.2	FAVAR and BFAVAR	42
2.2.3	Forecast evaluation	43
2.3	Data	43
2.4	Empirical Results	45
2.5	Conclusion	49
3	Comparing Proportions: A Modern Solution to a Classical Problem	59
	<i>José M. Bernardo</i>	
3.1	Introduction	59
3.2	Integrated Bayesian Analysis	60
3.2.1	Bayesian inference summaries	60
3.2.1.1	Point estimation	61
3.2.1.2	Region estimation	62
3.2.1.3	Precise hypothesis testing	62
3.2.2	Continuous invariant loss functions	63
3.2.3	The intrinsic logarithmic loss	65
3.3	Intrinsic Reference Analysis	67
3.3.1	Intrinsic reference estimation and testing	67
3.3.2	Example: The normal variance	68
3.4	Comparing Binomial Proportions	70
3.4.1	Intrinsic logarithmic loss to test equality	71
3.4.2	Reference posterior distributions	72
3.4.3	The RV144 HIV vaccine efficacy trial in Thailand	73
3.5	Discussion	77
4	Hamiltonian Monte Carlo for Hierarchical Models	79
	<i>Michael Betancourt and Mark Girolami</i>	
4.1	Hierarchical Models	80
4.2	Common Implementations of Hierarchical Models	82
4.2.1	Naïve implementations	83
4.2.2	Efficient implementations	83
4.3	Hamiltonian Monte Carlo for Hierarchical Models	85
4.3.1	Euclidean Hamiltonian Monte Carlo	87
4.3.1.1	Characteristic length scale	87
4.3.1.2	Limited density variations	88
4.3.2	Riemannian Hamiltonian Monte Carlo	89
4.4	Example	92
4.5	Conclusion	95

4.6	Appendix: Stan Models	97
5	On Bayesian Spatio-Temporal Modeling of Oceanographic Climate Characteristics	103
	<i>Madhuchhanda Bhattacharjee and Snigdhasu Chatterjee</i>	
5.1	Introduction	103
5.2	A Description of the Dataset	105
5.3	A Description of the Methodology	105
5.4	Results	108
5.5	Discussion	114
6	Sequential Bayesian Inference for Dynamic State Space Model Parameters	123
	<i>Arnab Bhattacharya and Simon Wilson</i>	
6.1	Introduction	123
6.2	Principle	125
6.3	Predicting and Filtering Density Approximations	126
6.3.1	Basic approximations	127
6.4	Examples: Linear Dynamic Model	127
6.5	Concluding Remarks	131
7	Bayesian Active Contours with Affine-Invariant Elastic Shape Prior	135
	<i>Darshan Bryner and Anuj Srivastava</i>	
7.1	Introduction	135
7.1.1	Past work on prior-driven active contours	136
7.1.2	Our approach and contributions	137
7.2	Affine-Invariant, Elastic Shape Statistics	138
7.2.1	Affine-invariant, elastic shape analysis	138
7.2.2	Statistical modeling	139
7.3	Bayesian Active Contour Model	143
7.3.1	Model definition	143
7.3.2	Computing E_{image} and its gradient	144
7.3.3	Computing E_{smooth} and its gradient	144
7.3.4	Computing E_{prior} and its gradient	145
7.4	Experimental Results	146
7.4.1	Multiview curve database (MCD)	147
7.4.2	Leaf segmentation	149
7.4.3	SAS shadow segmentation	150
7.5	Conclusion	153
8	Bayesian Semiparametric Longitudinal Data Modeling Using NI Densities	157
	<i>Luis M. Castro, Victor H. Lachos, Diana M. Galvis, and Dipankar Bandyopadhyay</i>	
8.1	Introduction	158

8.2	Semiparametric Linear Mixed-Effects Model	159
8.2.1	Normal/independent distributions	159
8.2.2	Model specification	160
8.3	Bayesian Approach	161
8.3.1	Modeling the non-parametric part	161
8.3.2	Prior distributions	163
8.3.3	Model comparison criteria	164
8.3.4	Case influence diagnostics	164
8.4	Application: ACTG 315 Data Revisited	165
8.5	Simulation Studies	167
8.6	Conclusions	174
8.7	Appendix	175
8.7.1	Appendix A: Densities of some specific NI distributions	175
8.7.2	Appendix B: Conditional posterior distributions	176
9	Bayesian Factor Analysis Based on Concentration	181
	<i>Yun Cao, Michael Evans, and Irwin Guttman</i>	
9.1	Introduction	181
9.2	The Model, Prior and Posterior	183
9.3	Concentration for Factor Analysis Models	188
9.4	Inferences and Examples	190
9.5	Conclusions	197
9.6	Appendix	197
10	Regional Fertility Data Analysis: A Small Area Bayesian Approach	203
	<i>Eduardo A. Castro, Zhen Zhang, Arnab Bhattacharjee, José M. Martins, and Tapabrata Maiti</i>	
10.1	Introduction	204
10.2	Theoretical and Methodological Background	205
10.2.1	Smoothing techniques applied to demographic analysis	205
10.2.2	Estimation of fertility rates for Portuguese NUTS III regions	207
10.3	Spatial Random-Effects Poisson Model	208
10.3.1	Model specification	208
10.3.2	Implementation	209
10.3.3	Model comparisons	210
10.4	Spatial Clustering Model for Estimating the Adjacency Matrix	210
10.4.1	Model specification	211
10.4.2	Implementation	212
10.4.3	Posterior inference	215
10.5	Application and Posterior Inference	215
10.6	Final Results and Conclusions	217

11 In Search of Optimal Objective Priors for Model Selection and Estimation	225
<i>Jyotishka Datta and Jayanta K. Ghosh</i>	
11.1 Introduction	225
11.2 The Optimal Prior	227
11.3 Shrinkage Priors	228
11.4 The Lasso Based Priors	229
11.5 Numerical Results	232
11.5.1 Effect of the irrepresentability condition	232
11.5.2 Performance comparison	232
11.5.3 Variable selection via inclusion probability	234
11.5.4 Estimation accuracy	235
11.5.5 Out-of-sample prediction error	236
11.6 Examples with Real Data	237
11.6.1 US crime data	237
11.7 Discussion	239
12 Bayesian Variable Selection for Predictively Optimal Regression	245
<i>Tanujit Dey and Ernest Fokoué</i>	
12.1 Introduction	245
12.2 Approaches to Response Prediction in Regression	247
12.2.1 Basic elements and assumptions	247
12.2.2 Methods of response prediction	250
12.3 Bayesian Approaches to Variable Selection	253
12.3.1 Bayesian variable selection with Zellner's g -prior	254
12.3.2 Spike and slab approach to variable selection	256
12.4 Computational Demonstrations	257
12.4.1 Analysis of the Diabetes dataset	257
12.4.2 Analysis of the Boston housing dataset	258
12.4.3 Analysis of the Prostate cancer dataset	258
12.4.4 Analysis of the Prestige dataset	259
12.4.5 Analysis of the Cars dataset	260
12.4.6 Analysis of the Hald Cement dataset	260
12.4.7 Analysis of the Attitude dataset	261
12.4.8 Simulated example with interactions	261
12.4.9 Final predictive comparison tables	262
12.5 Conclusion and Discussion	263
13 Scalable Subspace Clustering with Application to Motion Segmentation	267
<i>Liangjing Ding and Adrian Barbu</i>	
13.1 Introduction	267
13.2 Subspace Clustering by Spectral Clustering	269
13.3 Scalable Subspace Clustering by Swendsen–Wang Cuts	270

13.3.1	Posterior probability	271
13.3.2	Overview of the Swendsen–Wang Cuts algorithm	271
13.3.3	Graph construction	273
13.3.4	Optimization by simulated annealing	274
13.3.5	Complexity analysis	275
13.4	Application: Motion Segmentation	276
13.4.1	Dimension reduction	277
13.4.2	Experiments on the Hopkins 155 dataset	277
13.4.3	Scalability experiments on large data	281
13.5	Conclusion	283
14	Bayesian Inference for Logistic Regression Models Using Sequential Posterior Simulation	287
	<i>John Geweke, Garland Durham, and Huaxin Xu</i>	
14.1	Introduction	288
14.2	Sequential Posterior Simulation Algorithms for Bayesian Inference	289
14.2.1	Notation and conditions	289
14.2.2	Non-Adaptive SPS algorithm	292
14.2.3	Adaptive SPS algorithm	293
14.2.4	The two-pass SPS algorithm	295
14.3	Models, Data and Software	296
14.3.1	The multinomial logit model	296
14.3.2	Data	298
14.3.3	Hardware and software	300
14.4	Performance of the SPS Algorithm	300
14.5	Comparison of Algorithms	302
14.5.1	The exercise	302
14.5.2	Reliability	303
14.5.3	Computational efficiency	304
14.6	Conclusion	307
15	From Risk Analysis to Adversarial Risk Analysis	311
	<i>David Ríos Insua, Javier Cano, Michael Pellot, and Ricardo Ortega</i>	
15.1	Introduction	312
15.2	A Framework for Risk Analysis	312
15.3	Case Study: Fighting Fare Evasion in a Facility	314
15.3.1	Case description	314
15.3.2	Case modelling	315
15.3.3	Numerical results	317
15.4	A Framework for Adversarial Risk Analysis	321
15.5	Case Study: Fighting Fare Colluders in a Facility	323
15.5.1	The operator’s problem	324
15.5.2	The colluders’ problem	325
15.5.3	Numerical results	327

15.6 A Framework for Risk Analysis and Adversarial Risk Analysis	329
15.7 Case Study: Fighting Standard Evaders and Colluders Simultaneously	329
15.7.1 The operator's problem	330
15.7.2 The colluders' problem	331
15.7.3 Numerical results	331
15.8 Discussion	332
16 Symmetric Power Link with Ordinal Response Model	335
<i>Xun Jiang and Dipak K. Dey</i>	
16.1 Introduction	335
16.2 Model Setup for Ordinal Response Data	337
16.3 Ordinal Response Model with Symmetric Power Logit Link	338
16.4 Bayesian Computation	338
16.5 Data Analysis	340
16.6 Cross Validation	343
16.7 Conclusions	345
17 Elastic Prior Shape Models of 3D Objects for Bayesian Image Analysis	347
<i>Sebastian Kurtek and Qian Xie</i>	
17.1 Introduction	347
17.1.1 Related work	350
17.1.2 Utility of shape models in Bayesian image analysis	352
17.2 Representation of Surfaces	353
17.2.1 Pre-shape and shape spaces	354
17.2.2 Shape comparisons via geodesics	355
17.3 Shape Statistics of Surfaces	358
17.3.1 Estimation of the Karcher mean	359
17.3.2 Estimation of the Karcher covariance	359
17.3.3 Random sampling from Gaussian model	363
17.4 Summary	364
18 Multi-State Models for Disease Natural History	367
<i>Amy E. Laird, Rebecca A. Hubbard, and Lurdes Y. T. Inoue</i>	
18.1 Introduction	367
18.2 Multi-State Models: Background and Notation	368
18.2.1 Markov processes	369
18.2.2 Markov renewal processes and semi-Markov processes	370
18.3 Estimation and Inference: A Review	372
18.3.1 Methods for continuously observed processes	372
18.3.2 Panel data: Markov models	374
18.3.3 Panel data: Progressive semi-Markov models	374
18.3.4 Panel data: General semi-Markov models	377
18.3.5 Bayesian approach to multi-state models	380

18.4 Bayesian Multi-State Models: Applications	381
18.4.1 Nonhomogeneous Markov models: Modeling delirium	381
18.4.2 Semi-Markov models: Modeling HIV progression	383
18.5 Discussion	387
19 Priors on Hypergraphical Models via Simplicial Complexes	391
<i>Simón Lunagómez, Sayan Mukherjee, and Robert Wolpert</i>	
19.1 Introduction	392
19.1.1 Graphical models	393
19.2 Geometric Graphs	395
19.2.1 Example	397
19.3 Random Geometric Graphs	398
19.3.1 Simulation study of subgraph counts for RGGs	399
19.4 Model Specification	399
19.4.1 General setting	399
19.4.2 Prior specification	400
19.4.3 Sampling from prior and posterior distributions	401
19.4.3.1 Prior sampling	402
19.4.3.2 Posterior sampling	402
19.4.4 Convergence of the Markov chain	403
19.5 Three Simulation Examples	403
19.5.1 Example 1: $\mathcal{G}$ is in the space generated by $\mathcal{A}$	404
19.5.2 Example 2: Gaussian graphical model	405
19.5.3 Example 3: Factorization based on nerves	407
19.6 Discussion	409
20 A Bayesian Uncertainty Analysis for Nonignorable Nonresponse	415
<i>Balgobin Nandram and Namkyo Woo</i>	
20.1 Introduction	415
20.2 The Nonignorable Nonresponse Model	418
20.2.1 Projection strategy	420
20.2.2 Pooling strategy	421
20.3 Illustrative Example Using the Slovenian Plebiscite Data	422
20.3.1 Slovenian plebiscite data	422
20.3.2 Bayesian uncertainty analysis	424
20.4 Concluding Remarks	427
20.5 Appendix	428
20.5.1 Appendix A: Conditional posterior distributions	428
20.5.2 Appendix B: Inference for the finite population proportion	431

21 Stochastic Volatility and Realized Stochastic Volatility Models	435
<i>Yasuhiro Omori and Toshiaki Watanabe</i>	
21.1 Introduction	435
21.2 Stochastic Volatility Model and Efficient Sampler	436
21.2.1 Generation of $\theta = (\phi, \mu, \sigma_\eta^2, \rho)'$	438
21.2.2 Generation of $\mathbf{h}$: Single-move sampler	440
21.2.3 Block sampler for $\mathbf{h}$	440
21.3 Realized SV Model	445
21.3.1 Realized volatility	445
21.3.2 Realized SV model: Definition	446
21.3.3 Bayesian estimation using MCMC method	447
21.4 Empirical Application	448
21.5 Conclusion	453
22 Monte Carlo Methods and Zero Variance Principle	457
<i>Theodore Papamarkou, Antonietta Mira, and Mark Girolami</i>	
22.1 Introduction	458
22.2 Brief Overview of Variance Reduction in MCMC	459
22.3 Zero Variance Methodology	460
22.3.1 Choice of Hamiltonian operator	461
22.3.2 Polynomial trial functions	461
22.3.3 ZV in Langevin Monte Carlo	463
22.4 Quantification of Variance Reduction	465
22.5 Example 1; Normal Target	466
22.5.1 Univariate normal target	466
22.5.1.1 ZV with linear polynomial (LZV)	466
22.5.1.2 ZV with quadratic polynomial (QZV)	467
22.5.2 Multivariate normal target	469
22.6 Example 2; Multivariate t-Distribution Target	470
22.6.1 ZV with increasing degrees of freedom	471
22.6.2 ZV with increasing dimensionality	472
22.6.3 ZV with strongly correlated parameters	472
22.7 Discussion	473
23 A Flexible Class of Reduced Rank Spatial Models for Large Non-Gaussian Dataset	477
<i>Rajib Paul, Casey M. Jelsema, and Kwok Wai Lau</i>	
23.1 Introduction	477
23.2 Flexible Reduced Rank Spatial Model	480
23.2.1 Proposed FRRSM	483
23.3 MCMC Algorithm and Its Implementation	486
23.4 Application: Simulated and Real Data	491
23.4.1 Simulation study	491

23.4.2	Application to daily maximum of total column Ozone data	493
23.5	Conclusions and Discussion	498
24	A Bayesian Reweighting Technique for Small Area Estimation	503
	<i>Azizur Rahman and Satyanshu K. Upadhyay</i>	
24.1	Introduction	503
24.2	The Basic Concept	506
24.2.1	The Bayesian prediction theory	507
24.3	The Model	508
24.3.1	Matrix notations	508
24.3.2	The prior and posterior distributions	509
24.4	Modelling Unobserved Population Units	510
24.4.1	The linkage model	510
24.4.2	Predictive distribution	510
24.4.3	Joint posterior based reweighting	512
24.5	Empirical Results	513
24.5.1	Data source and computational process	513
24.5.2	Definition of housing stress	514
24.5.3	Estimates for private renters at small areas in Perth	514
24.6	Conclusion	516
25	Empirical Bayes Methods for the Transformed Gaussian Random Field Model with Additive Measurement Errors	521
	<i>Vivekananda Roy, Evangelos Evangelou, and Zhengyuan Zhu</i>	
25.1	Introduction	521
25.2	The Transformed Gaussian Model with Measurement Error	523
25.2.1	Model description	523
25.2.2	Posterior density and MCMC	525
25.3	Estimation of Transformation and Correlation Parameters	526
25.4	Spatial Prediction	528
25.5	Example: Swiss Rainfall Data	528
25.6	Discussion	534
26	Mixture Kalman Filters and Beyond	537
	<i>Saikat Saha, Gustaf Hendeby, and Fredrik Gustafsson</i>	
26.1	Introduction	538
26.1.1	Problem background	538
26.1.2	Organization	540
26.2	Particle Filtering	540
26.3	Rao-Blackwellization	541
26.4	Models for Mixture Kalman Filters	543
26.4.1	Hierarchical conditionally linear Gaussian model	543
26.4.2	Mixed linear/nonlinear Gaussian state-space model	544

26.5	Beyond Mixture Kalman Filter in Practice	547
26.5.1	Hierarchical conditionally finite HMM	547
26.5.2	Multiple switching nonlinear dynamic models	548
26.5.2.1	An RBPF implementation	548
26.5.3	Partially observed Gaussian state-space models	550
26.5.4	Partially observed finite HMM	552
26.5.5	Conditionally conjugate latent process model	552
26.5.5.1	Rao-Blackwellized noise adaptive PF	552
26.6	Terrain Navigation Problems	556
26.7	Approximate Rao-Blackwellized Nonlinear Filtering	557
26.8	Concluding Remarks	558
27	Some Aspects of Bayesian Inference in Skewed Mixed Logistic Regression Models	563
	<i>Cristiano C. Santos and Rosangela H. Loschi</i>	
27.1	Introduction	563
27.2	Logistic Regression Model with Random Intercept	565
27.2.1	Parameter interpretation and Larsen et al.'s results	566
27.2.2	Parameter interpretation under skew-normality	568
27.3	Bayesian Inference in Mixed Logistic Regression Models	570
27.3.1	The posterior full conditional distributions	570
27.3.2	Model selection	572
27.4	Monte Carlo Study	573
27.4.1	On the parameter estimates	573
27.4.2	On the random effects estimates	576
27.4.3	On the model selection	578
27.5	Final Comments	581
28	A Bayesian Analysis of the Solar Cycle Using Multiple Proxy Variables	585
	<i>David C. Stenning, David A. van Dyk, Yaming Yu, and Vinay Kashyap</i>	
28.1	Introduction	585
28.2	Modeling the Solar Cycle with Sunspot Numbers	589
28.2.1	Level one: Modeling the cycles	589
28.2.2	Level two: Relationships between consecutive cycles	591
28.2.3	Prior distributions	592
28.3	Incorporating Multiple Proxies of Solar Activity	593
28.3.1	Complete-data analysis	593
28.3.2	Multiple imputation strategy for missing data	594
28.4	Results	598
28.5	Summary and Discussion	603

29 Fuzzy Information, Likelihood, Bayes' Theorem, and Engineering Application	609
<i>Reinhard Viertl and Owat Sunanta</i>	
29.1 Introduction	609
29.2 Probability Based on Densities	610
29.3 Fuzzy Data and Likelihood Function	611
29.4 Generalized Bayes' Theorem	612
29.5 Example: Manual Lapping of Valve Discs	612
29.5.1 Surface finish (roughness)	613
29.5.2 Fuzzy Bayesian inference	614
29.6 Final Remark	617
30 Bayesian Parallel Computation for Intractable Likelihood Using Griddy-Gibbs Sampler	619
<i>Nuttanan Wichitaksorn and S.T. Boris Choy</i>	
30.1 Introduction	619
30.2 Parallel Computing and Griddy-Gibbs Sampler	621
30.2.1 Parallel computing basics	621
30.2.2 Griddy-Gibbs sampler	622
30.3 Simulation Study	623
30.3.1 Linear regression model with Student- t error	624
30.3.2 Nonlinear regression model	624
30.3.3 GARCH with asymmetric Laplace error	627
30.4 Empirical Study	628
30.5 Conclusion and Discussion	629
Index	633

List of Figures

1.1	Estimated changepoint probabilities for both regions using three different approaches to detrending and an independent noise model.	13
1.2	Estimated changepoint probabilities for both regions using three different approaches to detrending and a first order autoregressive noise model.	14
1.3	Modal model order shown as a function of space.	22
1.4	Posterior mean volume of distribution illustrated as a function of space.	22
1.5	SEF data. Static Model (left), Random Walk Model (centre) and dSPM estimate (right). Data presented on an inflated brain representation map.	28
2.1	Historical plots of the four major variables: 1997:04-2011:10.	44
2.2	Ex-ante forecast (2011:11 – 2012:10).	48
3.1	Intrinsic reference analysis for the standard deviation of a normal distribution.	70
3.2	Intrinsic logarithmic loss function.	72
3.3	Joint posterior reference density.	74
3.4	Reference posterior analysis of the vaccine efficacy for the RV144 vaccine efficacy trial data.	76
4.1	Example hierarchical model.	81
4.2	Spatially varying curvature in hierarchical models.	82
4.3	Slow exploration of hierarchical models with common Markov chain Monte Carlo algorithms.	84
4.4	Centered and non-centered parameterizations of hierarchical models.	85
4.5	Aggressive adaptation including biased inferences.	89
4.6	Limited variation of the log density of a hierarchical model using Euclidean Hamiltonian Monte Carlo.	90
4.7	Slow exploration of hierarchical models with Euclidean Hamiltonian Monte Carlo.	91
4.8	Performance of Euclidean Hamiltonian Monte Carlo with centered and non-centered parameterizations.	92

4.9	Robust tuning of Euclidean Hamiltonian Monte Carlo for parameterization experiments.	93
4.10	Large variation of the log density of a hierarchical model using Riemannian Hamiltonian Monte Carlo.	94
4.11	Rapid exploration of hierarchical models with Riemannian Hamiltonian Monte Carlo.	95
4.12	Robust tuning of Markov Chain Monte Carlo algorithms for performance comparisons.	96
5.1	Spatial location of data.	106
5.2	Posterior means at different longitudes and depths.	109
5.3	Posterior precisions at different longitudes and depths.	110
5.4	Posterior variances at different longitudes and depths.	111
5.5	Posterior mean of temperature at different longitude bands.	112
5.6	Posterior mean of salinity at different longitude bands.	113
5.7	Posterior mean of Oxygen-18 isotope ratio at different longitude bands.	114
5.8	The histograms depicting the posterior distribution of the μ parameters across <i>temperature</i> , <i>salinity</i> and <i>O-18 isotope ratio</i> .	115
5.9	Posterior distribution of the precision matrix elements.	116
5.10	Graphs denoting convergence properties, deviance, and autocorrelation structures in the MCMC runs.	117
6.1	Marginal posterior density of parameters using both INLA and SINLA.	129
6.2	Trace plots of mode and probability bounds for the unknown static parameters.	130
7.1	Shape models from top to bottom: [13], [6], [17], and our elastic affine-invariant model.	142
7.2	Segmentation results for occluded versions of the MCD shapes “crown” (camera angle 5) and “fountain” (camera angle 4).	149
7.3	Elastic shape models of the tulip poplar leaf. Top: Similarity-invariant. Bottom: Affine-invariant.	150
7.4	Segmentation results for tulip poplar leaves.	151
7.5	Example images from the PC SWAT cylinder database.	152
7.6	Elastic shape models of the SAS cylinder shadow. Top: Similarity-invariant. Bottom: Affine-invariant.	152
7.7	Segmentation results for the SAS database.	153
8.1	Viral load profiles and KL distance under SME model.	162
8.2	KL distance under NI-SME models.	168
8.3	J and L_1 distances under NI-SME models.	169
8.4	Simulated profiles and KL distances.	172
8.5	KL distances for simulated data.	174

9.1	The prior (—) and the posterior (– –) densities of the distance for $q = 0, 1$ and 2 in Example 3.	193
9.2	The prior (—) and the posterior (– –) densities of $d(\mathcal{R}, H_0^0)$ in Example 4.	194
9.3	The prior (—) and the posterior (– –) densities of $d(\mathcal{R}, H_0^1)$ in Example 4.	195
9.4	The prior (—) and the posterior (– –) densities of $d(\mathcal{R}, H_0^2)$ in Example 4.	195
9.5	Plots of prior (—) and posterior (– –) densities of distance to $\hat{\mathcal{R}}_2$ in Example 4.	196
10.1	Observed Fertility Rate by quinquennial age group j ($j = 1, \dots, 7$, ages 15 to 49 years old).	208
10.2	The central clustering configuration under SCP fit from 4,000 posterior samples.	216
10.3	Histogram of the posterior samples for SRP with estimated $\hat{\mathbf{W}}$ from SCPM.	217
10.4	Estimated Fertility Rate by quinquennial age group j ($j = 1, \dots, 7$, ages 15 to 49 years old).	218
10.5	Observed vs. Estimated Fertility Rate by quinquennial age group.	219
10.6	Observed vs. Estimated total number of births.	220
10.7	Observed vs. Estimated total fertility rates.	220
11.1	Probability density functions for generalized double Pareto, Cauchy, Laplace, Horseshoe and Robust prior.	230
11.2	Probability of selecting true model for Lasso, Robust prior and the Horseshoe prior.	233
11.3	Posterior inclusion probabilities due to the Robust prior and the Horseshoe prior.	235
11.4	Boxplots for the average squared error loss and the relative prediction error.	239
12.1	Boxplot of test errors for the Diabetes dataset with interactions.	258
12.2	Boxplot of test errors for the Boston Housing dataset.	259
12.3	Boxplot of test errors for the Prostate cancer dataset.	259
12.4	Boxplot of test errors for the Prestige dataset.	260
12.5	Boxplot of test errors for the Cars dataset.	260
12.6	Boxplot of test errors for the Hald Cement dataset.	261
12.7	Boxplot of test errors for the Attitude dataset.	261
12.8	Boxplot of test errors for the simulated dataset with interactions.	262
12.9	Boxplot of test errors for the simulated dataset with interactions and a correlation of $\rho = 0.72$.	263

13.1	Subspace clustering examples.	269
13.2	The spectral clustering algorithm [15].	270
13.3	The Swendsen–Wang Cut algorithm [2].	272
13.4	The Swendsen–Wang Cuts algorithm for subspace clustering.	274
13.5	Examples of SWC weighted graphs.	275
13.6	Sample images from some sequences of three categories in the Hopkins 155 database with ground truth superimposed.	278
13.7	Dependence of the misclassification rate on the affinity parameter m (left) and prior strength ρ (right).	278
13.8	SWC clustering example.	279
13.9	Selected frames of cars10 sequence with 1000 tracked feature points.	281
13.10	Scalability plot.	282
15.1	Risk analysis influence diagrams.	313
15.2	Influence diagram when only standard evaders are present.	315
15.3	Expected utility for security portfolios.	319
15.4	The Sequential Defend-Attack model.	321
15.5	Influence diagram when only colluders are present.	324
15.6	Fighting fare colluders.	324
15.7	Operator’s expected utility for colluders	328
15.8	Influence diagram for adversarial and non-adversarial threats.	330
15.9	Influence diagram when both types of evaders are present.	330
15.10	Operator’s problem when both types of evaders are present.	331
15.11	Operator’s expected utility for both evaders.	332
16.1	IPANE data collection locations.	336
16.2	Bar chart of coverage of <i>Berberis thunbergii</i> s.	340
16.3	Posterior density plot of power parameter r and shape parameter ξ in splogit and gev models.	342
17.1	Statistical shape analysis of surfaces should be invariant to translation, scale, rotation and re-parameterization.	348
17.2	Optimal re-parameterizations, also referred to as registration, provide improved shape comparisons and statistics.	350
17.3	Different representations of shapes of 3D objects.	351
17.4	Pictorial depiction of the mapping between the space of surfaces and the space of SRNFs.	355
17.5	Joint solution to the registration and optimal deformation/comparison problems.	356
17.6	Comparison of geodesics in the pre-shape space and shape space.	357
17.7	Comparison of geodesics in the pre-shape space and shape space. The bottom example considers two left putamen surfaces in a medical imaging application.	358

17.8	Comparison of sample means computed in the pre-shape space (a) and shape space (b).	360
17.9	Three principal directions of variation within ± 1 standard deviation of the mean for Examples 1 (top) and 2 (bottom) from Figure 17.8.	361
17.10	Principal direction of variation within ± 1 standard deviation of the mean for Examples 3 (top) and 4 (bottom) from Figure 17.8.	362
17.11	Four random samples from a generative Gaussian model in the pre-shape space (left panel) and the shape space (right panel).	363
18.1	Example of a J, X process, showing relationships between $\{(J_n, X_n), n \geq 0\}$ and associated process $Z(\cdot)$.	370
18.2	Examples of state models.	375
18.3	State model of [12].	377
18.4	State model of [19].	377
18.5	State models of [28].	378
18.6	State models of [7].	379
18.7	State model for delirium process.	381
18.8	Prior and posterior densities for transition intensities in time transformed random effects model for delirium data.	384
18.9	Estimated time transformation functions and 95% credible bands for stem cell transplantation patients with most extreme posterior median time transformation parameters from random effects time transformation model.	385
18.10	State model of [8].	385
18.11	Estimated cumulative distribution functions (CDFs) of the sojourn times in uninfected and infected states based on the data augmentation (DA) method and the method of DGL [8].	386
19.1	Proximity graph for 100 vertices and radius $r = 0.05$.	396
19.2	Top panel: (a) Alpha complex computed using $r = 0.5$ and the displayed vertex set. (b) The corresponding factorization.	397
19.3	Geometric graph representing the model given in Equation (19.6a).	404
19.4	Geometric graphs corresponding to snapshots of posterior samples from model of Equation (19.6a).	406
20.1	Boxplots of the posterior distributions of the finite population proportion as a function of the sensitivity bound	426
20.2	Plots of the estimated posterior densities of P , the hyperparameters, $\mu_0, \mu_1, \mu_2, \mu_3$ and, the sensitivity bound, B	427
21.1	Return and the posterior mean of log-volatility.	450

21.2	Daily return, log-realized kernel and the posterior mean of log-volatility.	451
23.1	The QQ plots of the detailed residuals: (a) for log-transformed data and (b) for untransformed data.	482
23.2	Simulation results for test data when using exponential, gamma, and GIG distributions for $\mathbf{W}$: (a), (c), (e): histograms of $Y - \hat{Y}$ and (b), (d), (f): scatterplot of Y vs. $\hat{Y}$.	492
23.3	Total ozone maximum in Dobson unit: (a) log transformed data, (b) data on original scale.	494
23.4	FRRSM results in log-scale: (a) predictions and (b) Monte Carlo standard errors.	495
23.5	BIW model in log-scale: (a) predictions and (b) Monte Carlo standard errors.	496
23.6	Maps of $Y - \hat{Y}$: (a) for FRRSM (b) for BIW model, and (c) location-wise $(MCSE^G - MCSE^W)/MCSE^G$.	497
23.7	FRRSM results in original scale: (a) predictions and (b) Monte Carlo standard errors.	498
24.1	A diagram of new system for reweighting small area micro-data.	506
25.1	Sampled locations for the rainfall example.	529
25.2	Contour plot of estimates of B_{γ, γ_1} for the rainfall dataset. The plot suggests $\hat{\lambda} = 0.1$ and $\hat{\phi} = 37$. Here the baseline value corresponds to $\lambda_1 = 0$ and $\phi_1 = 10$.	530
25.3	Estimates of B_{γ, γ_1} against λ for fixed value of $\phi = 37$ (left panel) and against ϕ for fixed values of λ (right panel). The circles show some of the skeleton points.	531
25.4	Maps of predictions (left panel) and prediction standard deviation (right panel) for the rainfall dataset.	532
25.5	Maps of predictions (left panel) and prediction standard deviation (right panel) for the rainfall dataset using the model of [5].	532
25.6	Histogram of random samples corresponding to the proportion of the area with rainfall larger than 200.	534
26.1	Mixture Kalman filter models.	544
26.2	Models with finite HMM substructure.	547
26.3	Models containing partially observable state space.	551
27.1	Prior distributions of OR	570
27.2	Prior distribution of λ under normal and skew-normal distributions.	574
27.3	MSEs for the posterior means for the fixed effects and the variance of the random effects.	575

27.4	Boxplots of posterior means for different scenarios.	577
27.5	Percentage of times each model provides the smallest and the second smallest MSE_{γ} .	579
27.6	Boxplots of MSE_{γ} for different scenarios.	580
28.1	The observed values of three proxies for solar activity: sunspot numbers, sunspot areas, and the 10.7cm flux.	588
28.2	A parameterized form for the solar cycle.	590
28.3	The Markov structure underlying a predictive model for the evolving solar cycle.	593
28.4	Transformations of three proxies for solar activity.	595
28.5	An illustration of the monotone missing data pattern of the solar proxies.	596
28.6	The fitted solar cycle.	599
28.7	Posterior predictive check of the model for the solar cycle.	600
28.8	Fitted values and 95% intervals for the rising times of the solar cycles.	601
28.9	Fitted values and 95% intervals for the falling times of the solar cycles.	601
28.10	Fitted values and 95% intervals for the total lengths of the solar cycles.	602
28.11	Predictions for solar cycle number 24.	602
29.1	Seat area of a valve disc.	613
29.2	A profile of surface roughness (R_a).	614
29.3	Fuzzy <i>a-priori</i> density.	615
29.4	Fuzzy sample.	615
29.5	Fuzzy <i>a-posteriori</i> density.	616
29.6	Fuzzy HPD interval for θ .	616
30.1	Convergence draws of linear regression model with Student- <i>t</i> error.	625
30.2	Posterior densities of linear regression model with Student- <i>t</i> error.	626

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List of Tables

2.1	One- to Twelve-Months-Ahead relative RMSE for Industrial production growth (2007:1-2011:10).	45
2.2	One- to Twelve-Months-Ahead relative RMSE for 3-month Treasury bill rate (2007:1-2011:10).	46
2.3	One- to Twelve-Months-Ahead relative RMSE for Indian Rupee-US Dollar nominal exchange rate (2007:1-2011:10).	46
2.4	One- to Twelve-Months-Ahead relative RMSE for Wholesale Price Index inflation rate (2007:1-2011:10).	47
4.1	Performance of Euclidean Hamiltonian Monte Carlo, Random Walk Metropolis, and Metropolis-within-Gibbs with centered and non-centered parameterizations.	97
6.1	Comparison of point estimates of the parameters, measures of accuracy and computation-time for SINLA, BSMC and INLA.	131
7.1	Top 5 singular values in the affine-invariant shape model from Figure 7.1.	143
7.2	Averaged segmentation results of occluded MCD shapes.	148
7.3	Segmentation results of the tulip poplar leaves.	151
8.1	Model selection criteria under NI-SME models.	166
8.2	Posterior estimates under NI-SME models.	167
8.3	Relative bias and MSE under NI-SME models.	170
8.4	Monte Carlo estimates of various model comparison measures.	171
8.5	Model selection criteria for simulated data.	173
8.6	Posterior estimates for simulated data.	173
9.1	The results of the iterative procedure to determine α_0 and β_0 in Example 1.	187
9.2	The true correlation matrix (above diagonal) in Example 3.	191
9.3	The estimated correlation matrix (above diagonal) in Example 3 when $q = 2$.	193
9.4	Sums of squared normalized importance sampling weights and coefficient of variation in Example 3.	193
9.5	Sample correlations in Example 4.	194
9.6	Estimated correlations in Example 4 based on H_0^2 .	196

10.1	Posterior inference for SRP, with posterior mean, (2.5%, 97.5%)th quantile from the total 3000 posterior samples.	217
11.1	Number of misclassified β_j 's for the Robust prior and the Horseshoe prior.	235
11.2	Comparison of average squared estimation error (and their standard error).	236
11.3	Comparison of average relative prediction error (and their standard error).	236
11.4	Description of the variables in US Crime dataset.	237
11.5	Description of the model selection methods used for the US Crime example.	238
11.6	Marginal Inclusion Probabilities for $p = 15$ variables in US Crime dataset.	238
12.1	Test errors over $R = 500$ hold-out subsamples	263
12.2	Test errors over $R = 500$ hold-out subsamples, with the training set representing 2/3 of the data.	264
13.1	Misclassification rates (in percent) of different motion segmentation algorithms on the Hopkins 155 dataset.	280
13.2	Average misclassification rate for the cars10 sequence (in percent).	282
14.1	Characteristics of datasets.	298
14.2	Reliability of one- and two-pass algorithms.	301
14.3	Log-marginal likelihoods, all datasets and models.	303
14.4	Posterior moments and numerical accuracy.	304
14.5	Log-marginal likelihoods and numerical accuracy.	305
14.6	Clock execution time in seconds.	305
14.7	Computational inefficiency relative to PSW.	306
15.1	Relevant features of counter measures.	314
15.2	Maximum planned investments.	318
15.3	Number of customers.	318
15.4	Expected inspections for each additional inspector.	319
15.5	Expected utilities for representative portfolios for different standard evaders scenarios.	320
15.6	Expected utilities for representative portfolios for different colluder scenarios.	328
15.7	Expected utilities when both types of evaders are present.	332
16.1	Posterior mean and standard deviation (SD) of parameters under different models.	341
16.2	Summary of percentage of difference between predicted response and observed response.	343

16.3	Observed counts and expected counts under different models.	344
16.4	Summary of training set and testing set for cross validation.	344
16.5	Summary of percentage of difference between predicted response and observed response for cross validation.	345
16.6	Observed counts and expected counts under different models for cross validation.	345
19.1	Summaries of the empirical distribution of edge and 3-clique counts for Čech complex random geometric graphs.	400
19.2	The three models with highest estimated posterior probability. The true model is shown in bold (see Figure 19.3).	405
19.3	The five models with highest estimated posterior probability. The true model is shown in bold.	407
19.4	Vertex set used for generating a factorization based on nerves.	408
19.5	Highest posterior factorizations with uniform prior for model of Equation (19.9).	409
19.6	Highest posterior factorizations with Strauss prior (true model is shown in bold).	409
20.1	Data from the Slovenian Public Opinion survey (SPOS).	423
20.2	Tables of observed and missing counts.	430
21.1	Descriptive statistics for daily returns.	448
21.2	Estimation result for asymmetric SV.	449
21.3	Estimation result for RSV.	451
21.4	Predictive ability of one-day-ahead volatility.	452
22.1	LZV estimators and their variances for the first three moments of a univariate normal target.	467
22.2	QZV estimators and their variances for the first three moments of a univariate normal target.	468
22.3	LZV estimators and their variances for the first three moments of a multivariate normal target.	470
22.4	VRFs for 2-dimensional t-distributions $t_\nu(\mathbf{0}, \frac{\nu-2}{\nu}I)$, $\nu = 5, 10, 25, 50, 100$.	472
22.5	VRFs for t-distribution of varying dimensions.	472
22.6	VRFs for 5-dimensional t-distributions $t_{25}(\mathbf{0}, \frac{23}{25}\Sigma(a))$ for $a = 0, 0.25, 0.5, 0.75, 0.99$.	473
24.1	Proportion of private renter households in housing stress at statistical local areas in Perth, Australia.	515
25.1	Posterior estimates of model parameters.	531
25.2	Coverage probability of one-sided prediction intervals $(-\infty, z_{0\alpha})$ for different values of α (first column) corresponding to our model (second column) and the model of [5] (third column).	533

27.1	Covariates and number of observations in each cluster.	574
27.2	Percentage of times each model provides the smallest MSE_{γ} .	578
27.3	Percentage of each model selection.	581
28.1	Checking the assumptions underlying the multiple imputation of the proxies for the solar cycle.	598
30.1	Estimation results of linear regression model with Student- t error.	626
30.2	Estimation results of nonlinear regression model.	627
30.3	Estimation results of GARCH(1,1) model with asymmetric Laplace error.	628
30.4	Estimation results of EGARCH(1,1) model using S&P 500.	629

Preface

The year 2013 can be considered an important year in the history of statistics, with six important global societies declaring the year as the International Year of Statistics. More than 128 countries participated in the International Year of Statistics. It added to its importance because 2013 was 300th anniversary of Jacob Bernoulli's *Ars conjectandi* and 250th anniversary of Bayes' theorem. India enthusiastically participated in this declaration by organizing an important Bayesian event in January, 2013 that may be considered the beginning of the celebrations of the International Year of Statistics in India. The 20th birth anniversary of International Society for Bayesian Analysis and 250th birth anniversary of the Bayes theorem in the presence of as many as 10 past, future and present presidents of ISBA was another important event celebrated in 2013 and simultaneously a highly desired publication in the form of the current volume *Current Trends in Bayesian Methodology with Applications* was decided to be brought out. We are happy that our thought has taken the shape of reality.

Bayesian statistics has expanded its coverage enormously in the past two to three decades. The Bayesian methods of reasoning are now applied to a wide variety of scientific, social, and business endeavours including areas such as astronomy, biology, economics, education, engineering, genetics, marketing, medicine, psychology, public health, sports, among many others. There are certain situations where Bayesian statistics appears as the only paradigm that offers viable solutions and this has become possible because of the tremendous development of Bayesian theory, methodology, computation and applications. The subject became the forefront of practical statistics with the advent of high-speed computers and sophisticated computational techniques especially in the form of Markov Chain Monte Carlo methods and sample based approaches. In fact, Bayesian modelling in complex problems freely combines components of different sorts of modelling approaches with structural prior information, unconstrained by whether such model combinations have ever been studied or analysed before.

Bayesian publications have also increased enormously in the last thirty years. Why then another publication on Bayesian statistics? The importance of the present volume may be realized from the fact that although the literature on Bayesian statistics is enormous, we do not find any single book that provides various aspects at one place. The literature is undoubtedly widely scattered, and journal articles rarely provide the conceptual background necessary for non-experts to understand and apply the approaches to their own

problems. Moreover, the cost of recent publications is getting so high that researchers find it difficult going through the massive amount of literature.

The present volume consists of thirty chapters. We have topics on biostatistics, econometrics, reliability and risk analysis, spatial statistics, image analysis, shape analysis, Bayesian computation, clustering, uncertainty assessment, applications to high-energy astrophysics, neural networking, fuzzy information, objective Bayesian methodologies, empirical Bayes methods, small area estimation, and a lot more. All the articles focus on Bayesian methodologies but each is self-contained and independent so that the present volume may lessen the competition with research level journals and may simultaneously act as a good reference for the researchers and graduate level students. We have preferred to include chapters giving an overview to the area including some theoretical insights and simultaneously expected to emphasize the work of others, give motivating examples, but omit sophisticated technical details. The present volume can be considered as a physical insignia of the inspiration by several experts in the field, which will be quite helpful for the future researches.

Though there may be topics closely related to each other, there is no need to maintain the sequence for reading the chapters. The sequence in the volume is purely alphabetical according to the last name of the first author and, in no way should be taken to mean any preferential order.

We fail in our duties if we do not express our sincere indebtedness to our referees, who were quite critical and unbiased in giving their opinion. We do realize that in spite of their busy schedules they offered us every support in timely commenting on various manuscripts. Undoubtedly, it is the joint endeavor of the contributors and the referees that emerged in the form of such an important and significant volume for the Bayesian world. We sincerely thank them all; the space constraint restricts us to mention the names individually.

We thankfully acknowledge the support rendered by John Kimmel, Executive Editor, Statistics, CRC Press, Taylor & Francis Group, who always stood behind us and always helped us with several of our unusual queries.

We express our indebtedness to everyone who was associated with us directly or indirectly while the work was in progress. The list is certainly too lengthy to be exhaustive but we would like to give special mention to Anuradha, Asha, Rita, Shakila, Geetika, Vertika, Debosri, Om Shankar, Naganandhini, Sangeetha, Rakesh, Rijji, Reema, Praveen, among others. At last but not the least, we express our thankfulness to Mr. Duvvuri Venu Gopal, Banaras Hindu University, who is credited with the present shape of the volume.

– The Editors

Foreword

It is a great pleasure to see a new book published on current aspects of Bayesian Analysis and coming out of India. This wide scope volume reflects very accurately on the present role of Bayesian Analysis in scientific inference, be it by statisticians, computer scientists or data analysts. Indeed, we have witnessed in the past decade a massive adoption of Bayesian techniques by users in need of statistical analyses, partly because it became easier to implement such techniques, partly because both the inclusion of prior beliefs and the production of a posterior distribution that provides a single filter for all inferential questions is a natural and intuitive way to process the latter. As reflected so nicely by the subtitle of Sharon McGrayne's *The Theory That Would Not Die*, the Bayesian approach to inference "cracked the Enigma code, hunted down Russian submarines" and more generally contributed to solve many real life or cognitive problems that did not seem to fit within the traditional patterns of a statistical model. Two hundred and fifty years after Bayes published his note, the field is more diverse than ever, as reflected by the range of topics covered by this new book, from the foundations (with objective Bayes developments) to the implementation by filters and simulation devices, to the new Bayesian methodology (regression and small areas, non-ignorable response and factor analysis), to a fantastic array of applications. This display reflects very well on the vitality and appeal of Bayesian Analysis. Furthermore, I note with great pleasure that the new book is edited by distinguished Indian Bayesians, India having always been a provider of fine and dedicated Bayesians. I thus warmly congratulate the editors for putting this exciting volume together and I offer my best wishes to readers about to appreciate the appeal and diversity of Bayesian Analysis.

– Christian P. Robert

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Bayesian Inference on the Brain: Bayesian Solutions to Selected Problems in Neuroimaging

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CONTENTS

1.1	Introduction	2
1.2	Background	3
1.2.1	Neuroscience and its imaging problems	3
1.2.2	Monte Carlo and sequential Monte Carlo	4
1.3	Application: Functional Magnetic Resonance Imaging	8
1.3.1	Problem	9
1.3.2	Methodology	10
1.3.3	Results	11
1.4	Application: Positron Emission Tomography	15
1.4.1	Problem	15
1.4.2	Background	16
1.4.3	Modelling & Methodology	18
1.4.3.1	Modelling	18
1.4.3.2	Methodology	19
1.4.4	Example	21
1.5	Application: Magnetoencephalography	23
1.5.1	Problem	23
1.5.2	Background	23
1.5.3	Modelling & Methodology	24
1.5.3.1	Modelling	24
1.5.3.2	Methodology	26
1.5.4	Example	27
1.6	Conclusions	29
	Acknowledgments	30
	Bibliography	31

1.1 Introduction

This chapter summarizes certain recent advances in inferential procedures for brain imaging. These advances involve the use of computationally intensive procedures. A goal of this chapter is to demonstrate that although the computational cost of these methods can be significant, it is feasible to employ them, using commodity-grade hardware, to provide Bayesian solutions to problems of realistic complexity in this domain.

Understanding the structure and the functional capabilities of the human brain is something that has been of profound interest for centuries. However, with recent advances in the ability to image the brain *in-vivo*, major advances in this understanding have been possible. Different neuroimaging (brain imaging) modalities typically have their strengths and weaknesses, usually consisting of trade-offs between spatial and temporal resolution and the invasiveness to the patient against the ability to probe specific or more general neuronal systems. However, the statistical methodologies that can be used to make inferences about differing systems can in many ways benefit from considering these modalities not in isolation but rather as inferential problems on spatio-temporal data sets.

It has long been recognized that (computational) Bayesian methods have the potential to provide excellent solutions to complex inferential problems (see, [71] for an application in reliability modelling, for example; and one need only consult the index of any recent volume on Bayesian inference, such as [70], to see how wide-reaching the influence of Monte Carlo methodology has been in this area). However, it is only recently that it has become feasible to apply these methods to the enormous datasets which arise in neuroscience and to do so sufficiently quickly that it is feasible to conduct the sensitivity analysis and replicated runs that are necessary to provide confidence in real applications. This chapter provides a non-technical introduction to these topics; more detailed explanations can be found within the references.

The subfield of neuroimaging is itself far too large to be exhaustively addressed here, and indeed even Bayesian neuroimaging has a very wide scope of applications including brain decoding [21] and more traditional group level hierarchical neuroimaging analyses [66]. Consequently, we confine ourselves here to three particular areas with which we have some experience and in which Bayesian methods provide good performance; there are of course a great many other problems within this field to which Bayesian techniques either have been or could be successfully applied.

1.2 Background

1.2.1 Neuroscience and its imaging problems

Neuroscience can be loosely defined as the study of the mechanism and function of the brain and nervous system, particularly within human subjects. Neuroimaging can be thought of as the subfield of neuroscience which is concerned with the acquisition of images of the (living) brain with a particular emphasis on images which attempt to illustrate the function of the brain and to show those regions which are related to the task or brain state of interest. It is often useful to obtain sequences of images which show, dynamically, changes in the brain during the course of some stimulus.

The focus of this article is the statistical analysis of data obtained in a variety of dynamic neuroimaging scenarios. The details of the different imaging techniques considered are described in the various application sections which follow, but they share a number of common elements. In all cases, complex indirect measurements of the brain's function are obtained.

In the field of neuroscience, Magnetic Resonance Imaging (MRI; [18]) is an imaging technique in which powerful radio-frequency magnetic fields are used to induce nuclear magnetic resonance within the nuclei of certain atoms within the brain. This resonance leads to a measurable perturbation in the magnetic field surrounding the subject. Functional Magnetic Resonance Imaging (fMRI; [34]) is particularly concerned with the location of oxygen-rich blood within the brain; which under the Blood Oxygen-Level Dependent (BOLD) hypothesis [53] is assumed to be related to regions of high neural activity. This technique is considered in Section 1.3.

Positron Emission Tomography (PET; [57]) is an invasive neuroimaging technique. Typically, a positron-labelled tracer is injected intravenously and the PET camera scans a record of positron emission as the tracer decays. With all events detected by the PET camera, the time course of the tissue concentrations are reconstructed as three-dimension images [38]. While it suffers from a significant disadvantage relative to fMRI, in that injection of radioactive material is required, it does benefit from being applicable to a whole range of neurochemical systems, rather than being restricted to the investigation of either direct or indirect changes of blood flow. The digital image so captured shows the signal integrated over small volume elements (voxels). The Bayesian estimation of model parameters of interest, together with selection or averaging is discussed in Section 1.4.

Magnetoencephalography (MEG; [32]) and Electroencephalography (EEG; [52]) are closely-related non-invasive neuroimaging techniques. Magnetoencephalography is an imaging technique which uses a helmet-shaped array of superconducting sensors to measure the magnetic fields produced by the neural currents in a human brain. The sampling rate of MEG recordings is typically around 1 kHz, which allows observation of dynamics on the millisecond

scale, several orders of magnitude faster than the two previously mentioned techniques of fMRI and PET. Among other non-invasive neuroimaging tools, only electroencephalography features a comparable temporal resolution. EEG is generally considered complementary to MEG, due to its different sensitivity to source orientation and depth [10], as it detects changes in electrical activity associated with neuronal responses as opposed to the changes in magnetic fields induced by electrical currents that are detected by MEG (see [52] for a comprehensive introduction to EEG). Note that estimation of the neural currents from the measured electric or magnetic fields is an ill-posed inverse problem [61] for which Bayesian inference provides a natural framework for regularization. We consider one particular task associated with inference for MEG in Section 1.5.

1.2.2 Monte Carlo and sequential Monte Carlo

In many Monte Carlo methods, an integral of interest is approximated by simulating a collection of random variables and using a (appropriately weighted) sample average to estimate an expectation. These procedures can be formally justified by appropriate versions of the law of large numbers and central limit theorem and are widely known and used in statistics. The basic principle is that the expectation of some function φ with respect to μ can be well approximated by the sample average $\frac{1}{n} \sum_{i=1}^n \varphi(x^i)$ where $x^1, \dots, x^n$ is a simple random sample from μ — or the realization of an ergodic μ -invariant Markov chain — and n is *large enough*. An excellent book-length summary is provided by [60].

We focus here on summarizing a less widely-used class of Monte Carlo methods which are used to address several of the neuroimaging problems addressed below. Sequential Monte Carlo (SMC) methods provide collections of weighted samples appropriate for approximating each of a sequence of distributions of interest in turn. Below, such methods are used to address the computational demands associated with Bayesian solutions to various problems in neuroimaging. These approaches are all based upon the framework of [14] which developed techniques for applying techniques developed in the setting of optimal filtering and dating back to [12, 30] (see [17] for a recent summary) to the approximation of arbitrary sequences of distributions and to providing good approximations of posterior distributions in particular.

SMC methods are based upon two important steps. The first of these is *importance sampling*, an elementary Monte Carlo technique almost as old as the idea of Monte Carlo itself (see, for example, [29]), that uses the identity

$$\int \varphi(x) \frac{\pi(x)}{\mu(x)} \mu(x) dx = \int \varphi(x) \pi(x) dx \quad (1.1)$$

which formally holds for any density μ which vanishes only where π itself vanishes. This allows expectations with respect to π to be approximated using samples from μ by computing a simple Monte Carlo estimate of the

expectation of $\varphi\pi/\mu$ using a sample from μ . This can be conveniently viewed as a weighted average with the *importance weight* π/μ correcting for the use of a different sampling distribution. Some simple conditions must be satisfied in order to ensure that the variance of this random approximation is finite, a simple strategy which guarantees this is to ensure that the tails of μ are heavier than those of π . See, for example, [26] for further details. In fact, one can use importance sampling in settings in which the importance weight, π/μ , is known only up to a normalising constant by estimating the normalising constant itself as the empirical mean of the unnormalised importance weights. This is important in Bayesian inference as the posterior density is typically known only to be proportional to the product of the prior and likelihood with the normalising constant, or evidence, being intractable.

The other key step within any SMC algorithm is *resampling*. The basis of this is that given a sample $x^1, \dots, x^n$ and associated weights $w^1, \dots, w^n$, which are normalised such that they sum to unity, one can stochastically replicate some of these samples and eliminate others in order to obtain an unweighted sample, $\bar{x}^1, \dots, \bar{x}^n$, such that:

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \varphi(\bar{X}^i) \middle| x^1, \dots, x^n, w^1, \dots, w^n \right] = \sum_{i=1}^n w^i \varphi(x^i), \quad (1.2)$$

provided only that the expected number of times that x_i is reproduced in $\bar{x}_1, \dots, \bar{x}_n$ is nw_i . See [15] for a comparison of some commonly-used resampling schemes.

Resampling in isolation can only increase the variance of the resulting estimate. Its use in SMC settings is due to the fact that it can stabilize the variance of estimates obtained after subsequent importance sampling steps within a framework known as *Sequential Importance Resampling (SIR)*. Given a sequence of distributions $\tilde{\pi}(\tilde{x}_n)$ on the spaces $\tilde{E}_n = \otimes_{p=1}^n E_p$ importance sampling can be carried out iteratively to provide weighted samples appropriate for targeting each target in turn.

More precisely, given a weighted sample suitable for approximating $\tilde{\pi}(\tilde{x}_n)$ one can first resample to obtain an unweighted sample, then extend each sample by sampling X_{n+1} from some Markov kernel, concatenating the existing and new samples to obtain samples $\tilde{x}_{1:n+1}$ on $\tilde{E}_{n+1}$ which can then be importance weighted to approximate $\tilde{\pi}_{n+1}$. Algorithm 1.1 shows the steps in a little more detail. Note that in this algorithm and all others in which simulated variables correspond to successive elements of a vector (as is usually the case when performing inference in time series models) we have introduced a double subscript with the first element corresponding to algorithm iteration and the second to the element within the vector. Whilst this may seem cumbersome it allows for cases in which subsequent algorithmic iterations alter the value of earlier elements of the state (and in even simple SMC algorithms, the resampling step in some sense does exactly this). It should be noted that numerous improvements to this algorithm are possible, especially via the use of adaptive

resampling schemes in which low variance resampling is used only in those iterations in which the weights have high variance; see [17] for a summary of such improvements.

Algorithm 1.1 The SIR Algorithm targetting $\{\tilde{\pi}_n\}_{n \geq 1}$

Iteration $n = 1$:

Sample $X_{1,1}^1, \dots, X_{1,1}^N \stackrel{\text{iid}}{\sim} \mu$

Calculate weights, **for** $i = 1, \dots, N$: $w(X_{1,1}^i) \propto \tilde{\pi}_1(X_{1,1}^i)/\mu(X_{1,1}^i)$.

Normalise weights, **for** $i = 1, \dots, N$: $W_1^i = w(X_{1,1}^i) / \sum_{j=1}^N w(X_{1,1}^j)$.

Iteration $n \geq 2$:

Resample $\{X_{n-1,1:n-1}^i, W_{n-1}^i\}_{i=1}^N$ to obtain $\{\bar{X}_{n-1,1:n-1}^i, 1/N\}_{i=1}^N$.

for $i = 1, \dots, N$: sample $X_{n,n}^i \sim q_n(\cdot | \bar{X}_{n-1,1:n-1}^i)$.

for $i = 1, \dots, N$: set $X_{n,1:n}^i = (\bar{X}_{n-1,1:n-1}^i, X_{n,n}^i)$.

Calculate weights, **for** $i = 1, \dots, N$:

$$w(X_{n,1:n}^i) \propto \tilde{\pi}_n(X_{n,1:n}^i) / \tilde{\pi}_{n-1}(X_{n-1,1:n-1}^i) q_n(X_{n,n}^i | X_{n-1,1:n-1}^i).$$

Normalise weights, **for** $i = 1, \dots, N$: $W_n^i = w(X_{n,1:n}^i) / \sum_{j=1}^N w(X_{n,1:n}^j)$.

Although the presentation thus far has been quite abstract, much of the development of SMC methods occurred in the field of time series analysis and in particular of online estimation within (general state space) Hidden Markov Models (HMMs; cf. [7] for a recent overview of these models and associated inference problems). HMMs are a broad class of time series models in which an unobserved Markov process $(X_n)_{n \in \mathbb{N}}$ evolves with some transition kernel $f(x_n | x_{n-1})$ and the observation process $(Y_n)_{n \in \mathbb{N}}$ leads to observations with the property that for any p , given x_p , Y_p is conditionally independent of the remainder of the hidden and observed processes and has density $g(y_n | x_n)$. That is, the joint density of $(X_{1:n}, Y_{1:n})$ is, for any n , simply:

$$p(x_{1:n}, y_{1:n}) = p(x_1) g(y_1 | x_1) \prod_{p=2}^n f(x_p | x_{p-1}) g(y_p | y_{p-1}).$$

Although Bayesian inference for this class of models, in particular the characterization of the posterior distribution of $X_{1:n}$ given $y_{1:n}$ may seem trivial, the unavailability for almost all interesting models of the marginal density of $Y_{1:n}$ means that one cannot compute the posterior density of $X_{1:n}$ analytically and is forced to resort to approximate methods. Noting that $p(x_{1:n} | y_{1:n}) \propto p(x_{1:n}, y_{1:n})$ allows us to apply Algorithm 1.1 directly to the sequence of distributions $\pi_n(x_{1:n}) := p(x_{1:n} | y_{1:n})$. Note that in this particular setting the algorithmic steps coincide with the arrival time of observations; when we deal with time series and this is not the case, we will reserve the symbol n to refer to algorithmic iterations and we will use t to refer to *real* time.

The simplest SMC algorithm applicable to filtering is the *bootstrap* particle filter of [30]. The basis of this approach is to set $q(x_n|x_{1:n-1}) = f(x_n|x_{n-1})$; in order to minimize the conditional variance of the importance weights whilst staying within the framework of Algorithm 1.1, [16] demonstrate that one should employ a locally optimal proposal of the form $q(x_n|x_{n-1}) \propto f(x_n|x_{n-1})g(y_n|x_n)$ (i.e. one should use a proposal which *conditions* upon the observed value of the current observation). The use of Markov chain Monte Carlo (MCMC) steps within SMC was first proposed by [27] who designed a filter in which MCMC was used to replace importance sampling; it's straightforward to combine the two in order to obtain the advantages of both techniques and an example of this is shown in the context of MEG in Section 1.5.

A number of approaches to extending the range of SMC methods have been developed in recent years [9, 45, 50] and to apply them to problems which do not have the structure described above. Most of these methods can be considered within the framework of [14], the key insight of which is that, given any sequence of target distributions $\pi_1, \dots, \pi_n$ on a sequence of spaces $E_1, \dots, E_n$, an auxiliary sequence of target distributions can be defined via:

$$\tilde{\pi}_n(\tilde{x}_{n,1:n}) = \pi_n(\tilde{x}_{n,n}) \prod_{p=1}^{n-1} L(\tilde{x}_{n,p+1}, \tilde{x}_{n,p}) \quad (1.3)$$

on the spaces $\tilde{E}_n = \otimes_{p=1}^n E_p$. This sequence of distributions trivially admits the target distribution of interest as a marginal distribution and is of the correct form to allow the application of SIR algorithms. In principle, the application of standard SIR algorithms to this sequence of auxiliary distributions will provide weighted samples appropriate for approximating the target distributions of interest. In practice, some care is needed over the design of these algorithms.

There has been some perception that these methods are difficult to implement, but it's our view that this perception is due more to a lack of familiar and, historically, of general software implementations than any real technical difficulties. In recent years, several software libraries have been made available to facilitate the implementation of SMC algorithms: simple and general C++ template libraries including [37] and [75], which makes parallel implementation very easy, and some more specialized packages which are likely to be easier to apply within their particular domains including [47]. It should be noted that the problems that will be introduced in the following sections naturally admit approaches based on SMC, either because they can be modelled explicitly or implicitly with HMMs or because the estimation of the normalising constant is of profound interest in the analysis (or because both are important). As will be shown, this will inherently allow similar SMC approaches to be used to investigate different neuroimaging modalities. Our aim is to demonstrate that Bayesian methods are able to provide good solutions to these problems and that SMC is able to facilitate these solutions in these settings; not to advocate this computational technique for every problem in neuroscience.

1.3 Application: Functional Magnetic Resonance Imaging

MRI is the application of nuclear magnetic resonance technology to the problem of imaging. In the context of neuroscience it is a widely used imaging technique in many contexts. In neuroimaging, the term “MRI” itself is synonymous with structural MRI, where anatomical maps of the brain are produced with the differing magnetic resonance (MR) signatures of grey matter, white matter and cerebrospinal fluid (CSF) enabling segmentation of the brain into differing anatomical structures. Another commonly usage of MRI is diffusion tensor MRI [40], which uses the preferential diffusivity of water along neuronal fibres as opposed to across the fibres to build structural maps of brain connectivity.

In this section, we will concentrate on functional MRI (fMRI). This use of MRI allows the detection of blood oxygenation levels with good spatial (and reasonable temporal) resolution; in combination with the BOLD hypothesis [53], the resulting image sequence can be used to infer neural activity. The BOLD hypothesis takes advantage of the differing magnetic properties of oxy- and deoxy- hemoglobin. As neuronal activity requires energy, changes in the relative concentrations of oxygen usage can be measured using fMRI, thanks to these differing magnetic properties. While this is essentially a measure of blood flow, subject to certain assumptions, this can be taken as a surrogate measure for the neuronal activity itself.

In most fMRI experiments, a known design is used in the experiment, allowing regression models (usually linear time series regression models) to be used for the estimation of parameters associated with the experimental task [74]. By investigating whether each spatial location is positively or negatively associated with the task, statistical parametric maps (as they are known in the neuroimaging literature) can be constructed for the task of interest. Correction methods are needed to account for the inherent multiple comparison issues associated with large numbers of tests (for a detailed overview of statistical methods commonly used in fMRI see [42]). However, in many situations, particularly those associated with psychological experiments, it is hard to be robust against the misspecification of the design, if specification is even possible. This is because reactions to psychological stimuli can typically evoke different responses from different subjects over different time scales. It has therefore been proposed [43] that a more natural framework for the investigation of such experiments might be unknown design regression, which can be framed as a change point problem. It is this approach that will be investigated further in this chapter.

A major issue in any modelling, and one which receives considerable discussion in the statistical analysis of fMRI data, is of course which statistical assumptions are appropriate to make when modelling the data. These include,

but are not limited to, the dependence or independence assumptions which are made, the presence of interesting or nuisance covariates and the stationarity or otherwise of time series. In this section, we will provide, through an fMRI example, an indicative investigation of how modifying these modelling assumptions can have a dramatic effect on the conclusions which can be drawn. This will be shown to be especially important in the context of change point analyses for fMRI time series.

1.3.1 Problem

In this section the particular problem of identifying both the number and location of *change points* within fMRI time series is considered. Given a sequence of fMRI observations, the observations associated with each region of the brain can be viewed as a time series. Three questions of interest then arise: how *many* qualitative changes in neural activity occur within each region, *when* do those changes occur, and *how certain* can we be about the answers to these questions.

A little more formally, let $\mathbf{y} = (y_1, y_2, \dots, y_t)$ denote the observations associated with a single neural region. Here, these observations are modelled as realising a (generalised) hidden Markov model with a small finite hidden state sequence $\mathbf{x} = (x_1, x_2, \dots, x_t)$ whose values correspond to a regime indicator. Change points in this data series are closely related to changes in the value of the hidden state. For a comprehensive treatment of HMMs, we refer the reader to [7].

A slightly broader class of models than those normally termed HMMs are considered, allowing some finite dependence on the past of both the observation and hidden processes. These models are characterised by the following conditional distributions:

$$\begin{aligned} Y_s | y_{1:s-1}, x_{1:s} &\sim f(y_s | x_{s-r:s}, y_{1:s-1}, \theta) && \text{(Emission)} \quad (1.4) \\ p(x_s | x_{1:s-1}, y_{1:s-1}, \theta) &= p(x_s | x_{s-1}, \theta) \quad s = 1, \dots, t && \text{(Transition).} \end{aligned}$$

Given the set of model parameters θ , the observation at time $s = 1, \dots, n$, y_s has emission density dependent on previous observations $y_{1:s-1}$ and previous r states of the underlying states $x_{s-r}, \dots, x_{s-1}$. The underlying states are assumed to follow a first order Markov chain (although this assumption can easily be relaxed) and takes values in a finite state space, E_X .

The simplest view of changepoints within this class of models, and one which is often used when working with HMMs, is that changepoints coincide exactly with changes in the value of the latent state. With this definition, saying that a changepoint occurs at time s is equivalent to stating that $x_{s-1} \neq x_s$.

We consider a slightly more general definition; a change point to a regime occurs at time s when the change in the underlying chain persists for at least k time periods. That is $x_{s-1} \neq x_s = \dots = x_{s+j}$ where $j \geq k-1$, and a change

of *regime* occurs only when there is a sustained movement in the underlying chain. This is done because spurious short segments will be of little interest in neuroimaging applications.

1.3.2 Methodology

By treating the problem parametrically within the family of generalised (finite state space) HMMs it is possible to compute, exactly, the posterior distribution over changepoint locations given the model parameters using the techniques of [3, 4]. These techniques rely upon Functional Markov Chain Imbedding (FMCI; [22]), and essentially consider an inhomogeneous Markov chain on an extended space which allows both the posterior distribution over changepoint location and the marginal likelihood $p(y|\theta)$, to be evaluated numerically using simple recursive algorithms.

In order to deal with parameter uncertainty, [48] employed an SMC sampler using a well established *likelihood tempering* strategy. This strategy uses a sequence of target distributions $\pi_n(\theta) \propto p(\theta)p(y_{1:t}|\theta)^{\gamma_n}$ where p denotes the prior distribution over the unknown parameters and γ_n is an increasing sequence with $\gamma_1 = 0$ and $\gamma_{n_{\max}} = 1$ which allows this sequence of artificial target distributions to interpolate smoothly between the prior and posterior distribution.

Algorithm 1.2 shows the steps of this algorithm in detail. In practice, some additional refinements are used, in particular resampling adaptively as required rather than every iteration (cf. [17]). Given the approximation of $p(\theta|y_{1:t})$ provided by this algorithm it is straightforward to approximate the distribution of the number of changepoints and their locations by writing the distribution of the quantity of interest as the marginal of the joint distribution of the quantity of interest and the unknown parameters and using the SMC approximation of the parameter posterior within that joint distribution.

We note that the use of essentially this approach, together with the natural evidence estimation procedure provided by the SMC algorithm, to perform model criticism, comparison or selection was further considered by [49]. Given the output of Algorithm 1.2 one can readily estimate the evidence using:

$$p(\widehat{y_{1:n}}|\theta) = \frac{1}{N} \sum_{i=1}^N \tilde{w}_1(\theta_0^i) \prod_{n=2}^{n_{\max}} \frac{1}{N} \sum_{i=1}^N \tilde{w}_n(\bar{\theta}_{n-1}^i, \theta_n^i). \quad (1.5)$$

It can be shown that this estimator is, in fact, unbiased [13]. Using estimates of this quantity obtained by running the algorithm for several competing models allows for straightforward Bayesian model comparison. In the context of determining how many regimes are required within a switching autoregressive model of the sort considered here, this strategy is advocated by [49].

One of the benefits of Bayesian methodology in this setting is that it provides a meaningful estimate of posterior uncertainty which is especially important in settings such as this one in which there can be many competing

Algorithm 1.2 SMC algorithm for quantifying the uncertainty in change points.

Approximating $p(\theta|y_{1:t})$

for $i = 1, \dots, N$ Sample $\theta_0^i \sim q_1$.

Compute for each i

$$W_1^i = \frac{w_1(\theta_0^i)}{\sum_{j=1}^N w_1(\theta_0^j)} \quad \text{where } w_1(\theta_0) = \frac{p(\theta_0)}{q_1(\theta_0)}$$

Resample $\{\theta_0^i, W_0^i\}_{i=1}^N$ to obtain $\{\bar{\theta}_0^i, 1/N\}_{i=1}^N$.

for each $i = 1, \dots, N$: Sample $\theta_1^i \sim K_1(\bar{\theta}_0^i, \cdot)$ where K_1 is a p -invariant Markov kernel.

for $n = 2, \dots, n_{\max}$ **do**

Reweighting:

for each $i = 1, \dots, N$ **compute:**

$$\tilde{w}_n(\bar{\theta}_{n-1}^i, \theta_n^i) = \frac{\pi_n(\bar{\theta}_{n-1}^i)}{\pi_{n-1}(\bar{\theta}_{n-1}^i)} = \frac{p(y_{1:t}|\bar{\theta}_{n-1}^i)^{\gamma_n}}{p(y_{1:t}|\bar{\theta}_{n-1}^i)^{\gamma_{n-1}}}.$$

and normalise $W_n^i = \tilde{w}_n(\theta_{n-1}^i, \theta_n^i) / \sum_{j=1}^N \tilde{w}_n(\theta_{n-1}^j, \theta_n^j)$.

Resample $\{\theta_{n-1}^i, W_n^i\}_{i=1}^N$ to obtain $\{\bar{\theta}_{n-1}^i, 1/N\}_{i=1}^N$.

for each $i = 1, \dots, N$: Sample $\theta_n^i \sim K_n(\bar{\theta}_{n-1}^i, \cdot)$ where K_n is a π_n invariant Markov kernel.

end for

Approximation of posterior:

$$p(\widehat{\theta|y_{1:n}})d\theta = \pi_{n_{\max}}(d\theta) \approx \sum_{i=1}^N \frac{1}{N} \delta_{\theta_{n_{\max}}^i}(d\theta).$$

explanations for the observed data which are all almost equally probable; simple maximization techniques fail to distinguish between this scenario and one in which a particular configuration of the latent states is overwhelmingly more probable than any alternatives.

1.3.3 Results

To illustrate the approach we consider a simple fMRI study of anxiety induction from [43] which was analyzed using the SMC approach described here in [48].

The experiment as described by [43]:

The design was an off-on-off design, with an anxiety-provoking speech preparation task occurring between lower-anxiety resting

periods. Participants were informed that they were to be given two minutes to prepare a seven-minute speech, and that the topic would be revealed to them during scanning. They were told that after the scanning session, they would deliver the speech to a panel of expert judges, though there was “a small chance” that they would be randomly selected not to give the speech.

After the start of fMRI acquisition, participants viewed a fixation cross for 2 min (resting baseline). At the end of this period, participants viewed an instruction slide for 15 s that described the speech topic, which was to speak about “why you are a good friend”. The slide instructed participants to be sure to prepare enough for the entire 7 min period. After 2 min of silent preparation, another instruction screen appeared (a “relief” instruction, 15 s duration) that informed participants that they would not have to give the speech. An additional 2 min period of resting baseline followed, which completed the functional run.

For brevity we consider the number of regimes to be known and equal to 2, but there would be little difficulty in simultaneously estimating the number of hidden regimes concurrently. We utilize a MS-AR(r) model of the form suggested by [55] for application to (suitably detrended) fMRI data:

$$y_t = \mu_{x_t} + a_t \quad a_t = \phi_1 a_{t-1} + \dots + \phi_r a_{t-r} + \epsilon_t \quad \epsilon_t \sim N(0, \sigma^2), \quad (1.6)$$

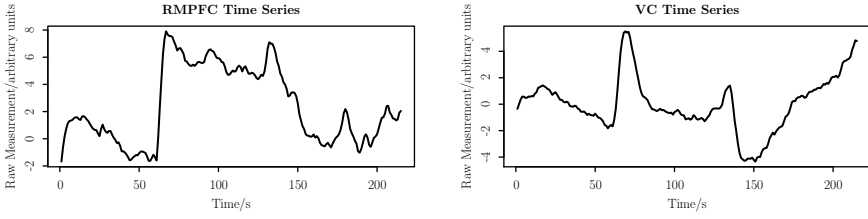
which allows an autoregressive dependence in addition to the latent variable structure.

Note that the a_t sequence is introduced only for notational convenience; the model can be written entirely in terms of the observation and latent state sequences, yielding for the emission density:

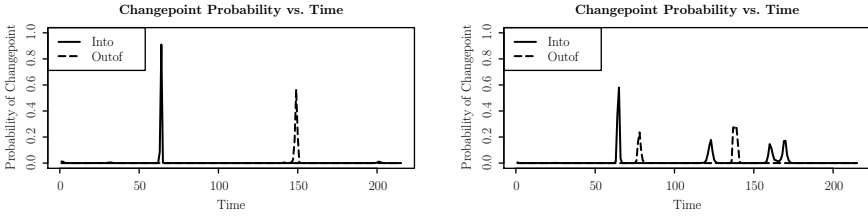
$$\begin{aligned} f(y_t | x_{1:t}, y_{1:t-1}, \theta) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2\sigma^2} \left(a_t - \left(\sum_{j=1}^r \phi_j a_{t-j} \right) \right)^2 \right) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2\sigma^2} \left((y_t - \mu_{x_t}) - \left(\sum_{j=1}^r \phi_j (y_{t-j} - \mu_{x_{t-j}}) \right) \right)^2 \right). \end{aligned} \quad (1.7)$$

Notice that the simple HMM with no autoregressive component is recovered when $r = 0$. A regime change is deemed to have occurred when the latent state persists for five consecutive time periods (i.e. $k = 5$ in the notation of subsection 1.3.1). This corresponds to typical estimates of haemodynamic response lengths [28].

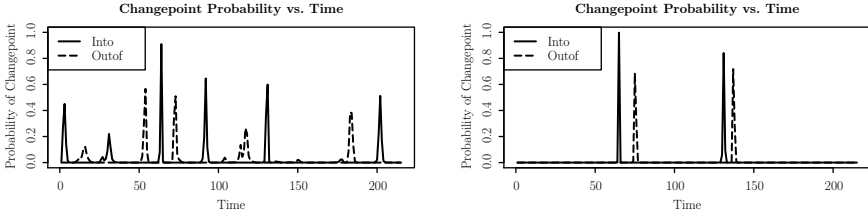
We consider two regions of the brain which are expected to experience significant activity during the course of this experiment: the rostral medial pre-frontal cortex (RMPFC), which is known to be associated with anxiety, and the visual cortex (VC), which is expected to show activity when the task-related instructions are seen, as indicated in [43].



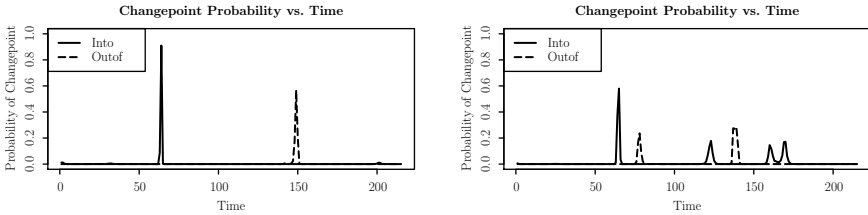
Unprocessed Data



Changepoint probabilities computed with no detrending.



Changepoint probabilities computed with DCB detrending.

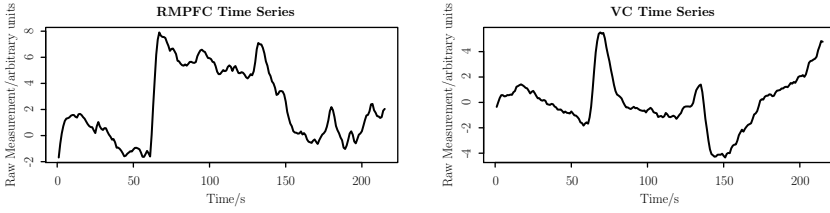


Changepoint probabilities computed with polynomial detrending.

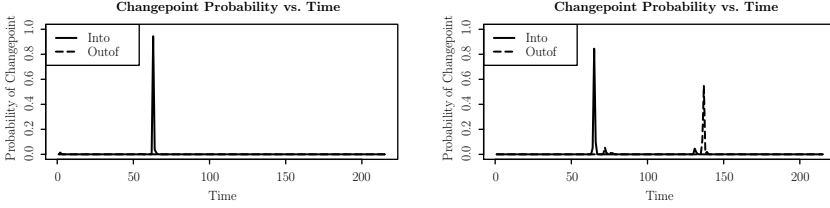
FIGURE 1.1

Estimated changepoint probabilities for both regions using three different approaches to detrending and an independent noise model.

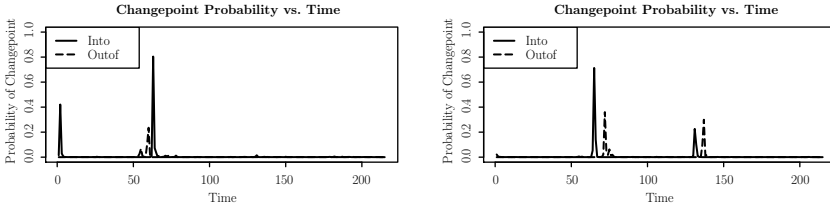
Changepoint probabilities are shown for each region using three different detrending schemes and an AR(0) (i.e., independent) noise model in Figure 1.1 and with an AR(1) noise model in Figure 1.2. In principle, detrending strategies and any other related preprocessing could also be automatically



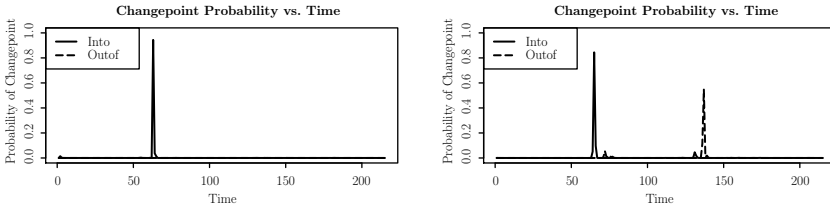
Unprocessed Data



Changepoint probabilities computed with nodetrending.



Changepoint probabilities computed with DCB detrending.



Changepoint probabilities computed with polynomial detrending.

FIGURE 1.2

Estimated changepoint probabilities for both regions using three different approaches to detrending and a first order autoregressive noise model.

selected within a Bayesian framework. However, prior specification in this case is non-trivial and due to confounding between the trend and possible change points, the best explanation of the data might not be a good strategy to consider when the preprocessing is potentially more flexible than the data model.

The detrending strategies are: to do no detrending and to analyze the raw data directly, to employ a low order (cubic) polynomial detrending [74] or to employ a discrete cosine basis (DCB) following [2]. These figures illustrate a number of things. They show that good estimation of the time of changes can be obtained *if* an appropriate detrending approach is considered and that the choice of detrending is appropriate. Change points present in time series can be confounded with slowly varying trends, and as such, care needs to be taken. The figures also show that results are strongly influenced by the choice of detrending procedure and, furthermore, that the use of an AR(1) noise model (with autoregressive parameter fixed at 0.20, as is the default behaviour of the SPM software [2]) can obscure much of the signal found in the data without this noise model.

It is clear that the findings of changepoint analysis in this setting are strongly influenced by preprocessing of the data and noise modelling and so it is extremely important to fully consider the implications of any modelling choices made and to assess the sensitivity of results to these choices when considering such an analysis. That said, the results obtained for the visual cortex with DCB detrending and for the RMPFC with no or polynomial detrending coincide very closely with neuro-physiological expectations illustrating the potential of even such simple models in this setting.

fMRI time series models are still routinely analyzed using known designs. However, hopefully, this chapter has shown that if care is taken in the choice of modelling parameters, it is feasible to determine the underlying design (along with associated uncertainty estimates) from the data rather than pre-specify it. While the analysis here has been undertaken only in a few regions, the increasing speed of algorithms, and the availability of massively parallel hardware offers the realistic possibility that real time change point detection in fMRI may be possible in the not too distant future.

1.4 Application: Positron Emission Tomography

PET is a widely used imaging technique which produces spatially-localised time course data for a very large number of voxels (typically of the order of 250,000). PET is invasive, requiring the introduction of a radionuclide-labelled tracer into the subject's bloodstream and so it is important to obtain the best possible inference from the limited quantity of data which is available.

1.4.1 Problem

There are a number of interesting problems to resolve when using PET to analyze neural activity. Here, we consider the estimation of a particular macroscopic quantity of interest, the *volume of distribution*; this quantity essentially

characterizes the quantity of tracer which would be found at a site in the brain in a steady-state regime obtained in the large time limit in a hypothetical experiment in which the tracer concentration in plasma is maintained at a constant level. In order to do this it is necessary to develop appropriate models for the process and, owing to the complexity of the problem at hand, to consider both parameter estimation and model selection/averaging.

1.4.2 Background

Compartmental models [35] are widely used in the modelling of PET. Such models describe the system of interest, in this case the dynamic behaviour of a radioactive tracer within a human brain, using a number of (real or conceptual) *compartments* between which a substance of interest, in this case the tracer, flows. Linear compartmental models are typically used in PET modelling for several reasons: the number of observations available is often very low and hence identifying more complex models can be difficult or impossible [62]; a single brain image typically contains at least 200,000 voxels and analysis needs to be conducted for every one of these voxels necessitating models for which inference can be performed extremely efficiently; and critically such models have been found to characterize PET data well [39].

Linear compartmental models can be conveniently characterised by a collection of linear differential equations: If we consider a linear m -compartment model and allow $f(t)$ to denote the vector whose i^{th} element corresponds to the concentration in the i^{th} compartment at time t . Let $b(t)$ describe all flow into the system from outside. The i^{th} element of $b(t)$ is the rate of inflow into the i^{th} compartment from the environment. The dynamics of such a model may be written as:

$$\frac{df}{dt}(s) = Af(s) + b(s), \quad f(0) = \xi, \quad (1.8)$$

where ξ is the vector of initial concentrations and the matrix A characterises the rate of flow between the various compartments (see [31]). The solution to this equation is,

$$f(t) = e^{At}\xi + \int_0^t e^{A(t-s)}b(s)ds,$$

where the matrix exponential $e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!}$.

It is possible to attach a loose qualitative interpretation to these equations, that there is an *inflow*, $b(t)$, into the system from outside and that material arriving in the system then flows between its compartments at rates depending upon A , and possibly also out of the system. This is a significant oversimplification in the case of the dynamics of a radioactive tracer entering the brain, and in some instances it's preferable to consider an alternative interpretation of the modelling. It's well known that compartmental models with sufficiently many compartments provide good approximation of essentially arbitrary dynamics. Indeed, in this context [46, p 200], observe that:

The linear combinations of exponential function forms have provided a very rich class of curves to fit to time-concentration data, and compartmental models turn out to be good approximations for many processes.

One can view the compartmental model as no more than a simple, pragmatic approximator of more complex dynamics — although some caution is needed as the small amount of data typically available severely limits the number of compartments which can be identified.

We consider only a particular class of compartmental models widely used for PET modelling, the *tracer input compartmental model*. In this case, it is assumed that in addition to the PET data, a measurement of the concentration of the tracer in plasma is available. The plasma measurement is generally treated as being noise-free as it can be made to a much higher degree of accuracy than the PET measurements themselves.

A plasma input model with m tissue compartments can be written as a set of differential equations,

$$\frac{dC_T}{dt}(s) = AC_T(s) + bC_P(s) \quad C_T(0) = \mathbf{0} \quad \bar{C}_T(s) = \mathbf{1}^T C_T(t),$$

where $C_T(t)$ is an m -vector of time-activity functions of each tissue compartment, $C_P(t)$ is the plasma time-activity function, i.e., the input function. A is the $m \times m$ state transition matrix, $b = (K_1, 0, \dots, 0)^T$ is an m -vector, where K_1 is the rate constant of input from the plasma into tissue. The vectors $\mathbf{1}$ and $\mathbf{0}$ correspond to the m -vectors of ones and zeroes, respectively. The matrix A takes the form of a negative semidefinite matrix with non-positive diagonal elements and non-negative off-diagonal elements. The solution to this set of ordinary differential equations (ODEs) is:

$$\bar{C}_T(t) = \int_0^t C_P(t-s)H_{TP}(s)ds \quad H_{TP}(t) = \sum_{i=1}^m \phi_i e^{-\theta_i t}, \quad (1.9)$$

where the ϕ_i and θ_i parameters are determined by the rate constants. The input function $C_P(t)$ is assumed to be nearly continuously measured. The tissue time-activity function $\bar{C}_T(t)$ is measured discretely, leading to measured values of the integral of the signal over each of n consecutive, non-overlapping time intervals ending at time points $t_1, \dots, t_n$.

The object of principal inferential interest, the aforementioned volume of distribution, is then:

$$V_D := \int_0^\infty H_{TP}(t)dt = \sum_{i=1}^m \frac{\phi_i}{\theta_i}.$$

In the next section a number of recent Bayesian approaches to estimation of the volume of distribution, together with the characterization of uncertainty, are described.

1.4.3 Modelling & Methodology

The traditional likelihood-based approaches to this problem are difficult to extend to allow for robust noise modelling and unknown model order.

A Bayesian approach to the estimation of the parameters of compartmental models of known model order was proposed in [56], who also provided a comparison with the state of the art in this area.

1.4.3.1 Modelling

In [76], a number of simple modelling improvements were made: considering non-normal noise distributions (much heavier tailed noise is strongly supported by the real data considered below) and developing biologically-motivated prior distributions for the parameters of interest. However, the main focus of that work was extending the approach of [56] to allow for the estimation of model evidence and hence to permit model selection or averaging. The strategy was based around MCMC together with a *careful* implementation of a generalised harmonic mean estimator.

Both [56] and subsequently [76] made use of a model of the form:

$$\bar{C}_T(t_j; \phi_{1:m}, \theta_{1:m}) = \sum_{i=1}^m \phi_i \int_0^{t_j} C_P(s) e^{-\theta_i(t_j-s)} ds$$

$$y_j = \bar{C}_T(t_j; \phi_{1:m}, \theta_{1:m}) + \sqrt{\frac{\bar{C}_T(t_j; \phi_{1:m}, \theta_{1:m})}{t_j - t_{j-1}}} \varepsilon_j,$$

where m denotes the number of tissue compartments (1, 2 or 3 in this work), $t_0 = 0$, and $\{\varepsilon_j\}$ are identically independently distributed random variables with mean zero.

In most works it has been assumed that the innovations, ε_j are normally distributed; [76] compares this assumption with one in which t distributions with more than two degrees of freedom are permitted and found that real PET data provided overwhelming support for a heavy-tailed noise distribution and so only that model is considered here. That is, the innovation terms are modelled using zero-mean t random variables, $\varepsilon_j \sim \mathcal{T}(0, \tau, \nu)$, where $\mathcal{T}(0, \tau, \nu)$ denotes the centered Student t distribution with scale τ , and ν degrees of freedom.

The prior distribution for scale τ is an inverse gamma distribution with both parameters equal to 0.001 — a proper approximation to the improper Jeffrey’s prior — while that for $1/\nu$ is uniform over interval $[0, 0.5)$, allowing the tails to the likelihood to vary between those of the normal and the much heavier tails of a t distribution with 2 degrees of freedom (following [24]).

Prior distributions for the rate parameters of the three compartment model were obtained from biophysical considerations by [76] (who also considered vague conjugate priors and demonstrated that parameter estimation and model selection results were somewhat insensitive to the precise choices of

prior distribution):

$$\begin{aligned}
 \phi_1 &\sim \mathcal{TN}_{[10^{-5}, 10^{-2}]}(\cdot; 3 \times 10^{-3}, 10^{-3}) \\
 \theta_1 | \phi_1 &\sim \mathcal{TN}_{[2 \times 10^{-4}, 10^{-2}]}(\cdot; \phi_1/15, 10^{-2}) \\
 \phi_2 &\sim \mathcal{TN}_{[10^{-5}, 10^{-2}]}(\cdot; 10^{-3}, 10^{-3}) \\
 \theta_2 | \phi_2, \theta_1 &\sim \mathcal{TN}_{[\theta_1, 6 \times 10^{-2}]}(\cdot; \phi_2/4, 10^{-2}) \\
 \phi_3 &\sim \mathcal{TN}_{[10^{-5}, 10^{-2}]}(\cdot; 10^{-3}, 10^{-3}) \\
 \theta_1 | \phi_3, \theta_2 &\sim \mathcal{TN}_{[\theta_2, 6 \times 10^{-2}]}(\cdot; \phi_3/1, 10^{-2}).
 \end{aligned}$$

where the density of the truncated normal distribution is

$$\mathcal{TN}_{[a,b]}(x; \mu, \sigma^2) := \frac{\mathcal{N}(x; \mu, \sigma^2) \mathbb{I}_{[a,b]}(x)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)},$$

with $\mathbb{I}_{[a,b]}$ corresponding to the indicator function on $[a, b]$ and Φ to the standard normal distribution function. For one- and two-compartment models the appropriate subset of these prior distributions are used, ensuring that common priors are used for the shared parameters of nested models.

1.4.3.2 Methodology

The generalised harmonic mean estimator (due to [23]; extending the harmonic mean estimator of [51]) is based around the simple identity, which holds for any probability density, $g(\theta)$:

$$\int_{\theta \in \Theta} g(\theta) \frac{p(\theta|y)p(y)}{p(y|\theta)p(\theta)} d\theta = 1 \Rightarrow p(y) = \left[\int_{\theta \in \Theta} p(\theta|y) \frac{g(\theta)}{p(y|\theta)p(\theta)} d\theta \right]^{-1}.$$

In principle, writing the quantity of interest as the reciprocal of a posterior expectation in this way allows the estimation of $p(y)$ using any sample approximation of the posterior, $\{\theta^i\}_{i=1}^N$ by simply computing:

$$\hat{p}(y) = \left[\frac{1}{N} \sum_{i=1}^N \frac{g(\theta^i)}{p(y|\theta^i)p(\theta^i)} \right]^{-1}.$$

However, this estimator is essentially an importance sampling estimator and will be well-behaved (and have finite variance) only if the tails of g are light compared with those of the true (and typically unknown) posterior density. It will have a small variance only if the dissimilarity between g and the posterior is small. This issue has been the subject of much discussion in the literature [59]. In this application it *is* possible to do this robustly but doing so comes at the cost of both a considerable implementation effort and the use of an adaptive scheme to tune the parameters of the MCMC algorithm as in [76].

The potential of SMC to provide efficient characterization of the posterior distribution over both model and parameter space in the PET setting

was noted by [77]; the approach proposed there corresponds essentially to Algorithm 2 of [78], with Algorithm 4 of that document describing a more sophisticated, adaptive variant of the same general approach which requires little user intervention. This simple algorithm is referred to as SMC2-DS in these papers (the 2 denoting which of three possible approaches to the problem is employed and the DS that the evidence is estimated using Equation 1.5 rather than via an approximation of the path-sampling estimator of [25] which is also possible).

Algorithm 1.3 An SMC approach to evidence estimation for PET compartmental models.

for each number of compartments, $k \in \{1, \dots, k_{\max}\}$:

Initialisation: Set $n \leftarrow 0$.

Sample $\theta_0^{(k,i)}, \dots, \theta_0^{(k,N)} \stackrel{\text{iid}}{\sim} p(\theta|M_k)$ from the parameter prior.

Set Weight $\{W_0^{(k,i)} = 1/N\}_{i=1}^N$.

Iteration: Set $n \leftarrow n + 1$.

for $i = 1, \dots, N$: Weight $W_n^{(k,i)} \propto W_{n-1}^{(k,i)} p(\mathbf{y}|\theta_{n-1}^{(k,i)}, M_k)^{\gamma_n^k - \gamma_{n-1}^k}$.

Apply resampling if necessary.

for $i = 1, \dots, N$: Sample $\theta_n^{(k,i)} \sim K_n(\cdot|\theta_{n-1}^{(k,i)})$, a $\pi_{k,n}$ -invariant kernel.

Repeat the Iteration step until $n = n_{\max}^k$.

end for

Use Equation 1.5 to estimate $p(\mathbf{y}|M_k)$.

To avoid dwelling unnecessarily on technical details, Algorithm 1.3 shows the simplest form of the algorithm. Here, $n_{\max}^k$ describes the number of distributions in the sequence used to deal with model k , which we denote M_k and which corresponds to the k -tissue compartment model in our setting. Tuning parameter γ_n^k corresponds to the n^{th} step in the tempering sequence for model k , i.e., $\pi_{k,n}(\theta) \propto p(\theta|M_k)p(\mathbf{y}|\theta, M_k)^{\gamma_n^k}$. Note that the sequence of distributions employed and the scaling of the proposal distributions can be automated by using the adaptive techniques described in [78] and that this is a major strength of the approach in problems such as that considered here in which it needs to be applied to hundreds of thousands of data series. The adaptation means that a common algorithm can be applied to *all* of these series without tuning on a case-by-case basis and it can then be robust enough to produce good estimation for all of those sequences. Indeed, it's not necessary to specify even the number of intermediate distributions used in each sequence as the adaptive scheme is able to select this (and hence the computational effort) with only one user-specified precision parameter being set. Another significant benefit of the SMC approach is that it can be readily parallelized and can take advantage of modern parallel architectures, see [41] for some illustrations, even those which depend upon SIMD (Same Instruction Multiple Data) operations (in the context of model selection this is studied in some depth by [78]).

1.4.4 Example

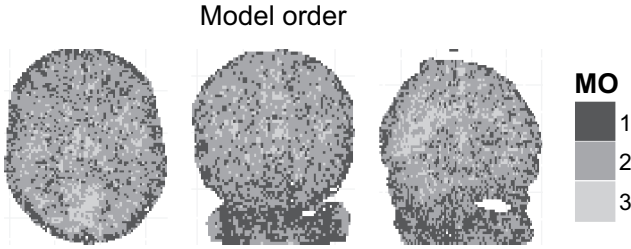
We consider data obtained from a PET study using [^{11}C]diprenorphine. The original intention of the study was to quantify opioid receptor concentration in the brain of healthy subjects allowing a baseline to be found for subsequent studies on diseases such as epilepsy which tend to involve changes in brain receptor concentrations or occupancy levels either due to physical lesions within the brain or other chemically relevant differences from controls.

These data have been previously analysed in [36, 56], who focussed on the parameter estimation problem, and by [77] who applied MCMC and generalised harmonic mean estimation to the problem. Another data set from the same study was considered by [78]. Dynamic scans from an [^{11}C]diprenorphine study of healthy subjects, for which an arterial input function was available, were analyzed. [^{11}C]diprenorphine is a tracer that binds to the opioid (pain) receptor system in the brain. The subjects underwent 95 min dynamic [^{11}C]diprenorphine PET baseline scans. Each patient was injected with [^{11}C]diprenorphine with a total activity of 185 MBq. PET scans were acquired in 3D mode on a Siemens/CTI ECAT EXACT3D PET camera, with a linear spatial resolution after image reconstruction of approximately 5 mm. Data reconstruction employed the reprojection algorithm [38] with ramp and Colsher filters cut off at the Nyquist frequency. Reconstructed voxels were $2.096\text{ mm} \times 2.096\text{ mm} \times 2.43\text{ mm}$. Acquisition was performed in listmode (event-by-event) and scans were rebinned into 32 time frames of increasing duration. Frame-by-frame movement correction was performed on the PET images. Overall this resulted in images of size $128 \times 128 \times 95$ voxels which, when masked to include only brain regions, resulted in $\sim 250,000$ separate time series to be analyzed. Each of these time series is analyzed separately within the Bayesian analysis considered here.

We provide here a short summary of the results of this approach while referring the reader to [76] for a comparison of Bayesian and alternative approaches to estimation and to [78] for further details of the sequential Monte Carlo approach and results of the analysis of another data set from the same study.

Figure 1.3 shows three orthogonal slices through the brain, each illustrating the posterior modal model order m obtained for each voxel using the adaptive SMC scheme described. The spatial distribution of the model order can give a crude indication of the types of biological processes occurring at different spatial locations. Figure 1.4 shows the volume of distribution estimated under the most probable model at the same locations. In real applications, it might be more appropriate to consider model averaging, rather than selection, for the purposes of computing parameters of interest; the selection approach is employed here only to simplify exposition.

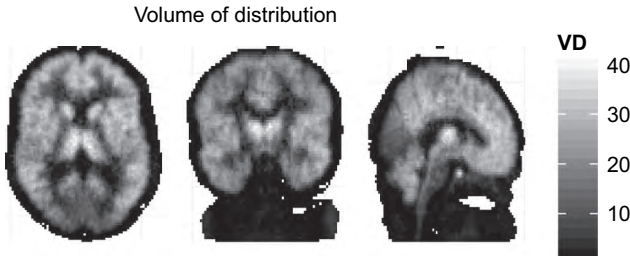
Both figures show considerable spatial structure: this is not an artefact of the modelling as each voxel was treated independently, but reflects information present within the data itself. There is a non-trivial relationship here

**FIGURE 1.3**

Modal model order shown as a function of space.

between $\mathbb{E}[V_D]$ and model order. Prior distributions are typically given for micro parameters such as those in the A matrix of (1.8), while the parameter of interest is typically V_D . However, priors on such parameters are much easier to interpret and are therefore the standard in Bayesian PET models. In addition, such specifications are unlikely to cause major issues in interpreting the posterior of V_D . We note, in passing, that similar structure is not readily apparent when traditional likelihood and information-criteria-based methods are employed [76]. Figure 1.4 shows clearly the physical structure of the brain and, perhaps more interestingly, provides a clear indication that the tracer is much more highly concentrated within certain regions of the brain (as would be expected physiologically).

The analysis in this chapter has demonstrated that massive data, in this case hundreds of thousands of voxels, can be analyzed using advanced Monte Carlo techniques. As imaging modalities such as PET become increasingly important in a whole range of clinical applications from cancer studies to psychiatric disorders, robust, automated, computationally tractable Bayesian approaches have the potential to make large contributions to the analysis of such studies.

**FIGURE 1.4**

Posterior mean volume of distribution illustrated as a function of space.

1.5 Application: Magnetoencephalography

Magnetoencephalography enjoys extremely good (~ 10 ms) temporal resolution and is often used in settings in which the temporal evolution of signals is of interest. The problem of determining the underlying pattern of neural currents from the collection of measured magnetic field strengths is ill-posed and regularized solution of this inverse problem has attracted a great deal of attention.

1.5.1 Problem

This section addresses the specific problem of estimating, online as observations become available, the posterior distribution over current neural currents given the observations received thus far. The approach involves the use of sequential Monte Carlo algorithms within a simple object tracking framework which attempts to make use of a model form the dynamics of the neural currents in order to provide the needed spatio-temporal regularization. Although the methodology discussed has not yet been implemented in real time, doing so is certainly within the range of current computational hardware.

1.5.2 Background

Numerous approaches to the problem of estimating the configuration of neural currents from MEG data have been developed. Broadly speaking, each method uses one of two classes of approximation of the neural currents: *distributed source* models in which currents are treated as varying smoothly throughout the brain or *current dipole* models in which a small number of infinitesimal current dipoles (the closest thing to a point-source of magnetic field which exists) are used to model the current distribution. Both approaches have advantages and disadvantages. See [73] for a detailed discussion.

The simplest approaches produce a single estimate of the *spatial location* of activity averaging over time. Common approaches include Minimum Norm Estimation (MNE; [33], a natural L2-regularized approach) and Minimum Current Estimation (MCE; [72], an L1-regularized approach). These regularized approaches invariably fall into the class of distributed source models.

More recently, approaches which treat the problem explicitly as a dynamic one have been developed. Spatiotemporal regularization approaches attempt to exploit the qualitative observation that activity is expected to be spatially sparse but temporally smooth, typically employing two different norms to provide appropriate regularization in each domain. Works taking such an approach include [54, 68].

Inference under a Bayesian dynamic model which fully describes the evolution of the system has become possible only with the availability of powerful

computers. [44] address exact computation of posterior densities under linear Gaussian modelling assumptions imposed upon a (discretised) distributed source model, but such an approach has extremely high computational requirements in addition to substantial modelling restrictions.

A number of papers have considered Monte Carlo approximation of the sequence of filtering distributions using current dipole models (i.e. the distributions over neural current configuration at each time instant given all of the observations received up until that time) using SMC. These include [63, 65] which propose basic particle filters and [6] which attempted to implement an approximation of a so-called Rao-Blackwellized particle filter [1, 8] in which a combination of analytical and Monte Carlo methods are combined by integrating out some unknown variables conditional upon simulated values of the others. In general, this is an excellent idea and one which can significantly reduce the Monte Carlo variance of the resulting estimator; unfortunately, in this case approximations were required to arrive at such an algorithm and these approximations are such that the resulting estimates would not recover the true posterior distribution as the Monte Carlo sample size tends to infinity. All these papers use a model in which current dipoles move around the brain according to a random walk; this is not consistent with the biophysical interpretation of these current dipoles but is required for computational reasons.

Recently, [64] proposed an algorithm which allows for consistent Monte Carlo approximation of the filtering distributions associated with a model in which current dipoles do not move. This approach is discussed in more detail in the following section.

1.5.3 Modelling & Methodology

Multiple object tracking is a difficult problem and has attracted a great deal of attention in the literature ([58] identifies 35 different *types* of multiple object tracking algorithm). The problem within the current dipole model of MEG image reconstruction is especially challenging due to the characteristics of the underlying dynamics.

1.5.3.1 Modelling

As with most tracking problems, the modelling is naturally decomposed into two components: one which describes the underlying dynamics of the hidden system being tracked (in this case, these describe the evolution of the current dipoles within the brain) and a second which describes the relationship between that hidden system and the measurement apparatus. This section commences with a description of the measurement model before returning to a discussion of the dynamic model.

In principle, the magnetic field arising from any latent neural current configuration can be computed directly via Maxwell's equations (or in a steady

state regime via the Biot-Savart law). However, doing so can be computationally costly and it is common practice to discretize the brain volume and to compute the influence of three orthogonal current vectors at the centre of each of the resulting N_{grid} volume elements offline in advance and stored in the *leadfield* matrix, G . The forward problem associated with any configuration of current elements located at the centre of those volume elements can then be solved (under the magnetostatic approximation, in which the influence of time varying currents is neglected) by a simple linear superposition.

Such an approximation gives rise to the following measurement model (note that throughout this section parenthetical superscripts index individual dipoles within a collection):

$$b_t = \sum_{i=1}^{N_t} G(r_t^{(i)}) q_t^{(i)} + \epsilon_t \quad (1.10)$$

where b_t is a vector of magnetic currents measured at N_{sensors} detectors at time t , N_t is the number of active dipoles at time t , $q_t^{(i)}$ denotes the i^{th} dipole moment at time t , $r_t^{(i)}$ is the location of dipole i at time t relative to each detector element, $G(r_t^{(i)})$ is the $N_{\text{sensors}} \times 3N_{\text{grid}}$ subset of the leadfield matrix describing the influence of a dipole at location $r_t^{(i)}$ on each sensor and ϵ_t is an additive noise vector.

Assuming that the ϵ_t are independent Gaussian random vectors with covariance Σ_{noise} leads to the likelihood:

$$p(b_t | j_t) = \mathcal{N} \left(b_t; \sum_{i=1}^{N_t} G(r_t^{(i)}) q_t^{(i)}, \Sigma_{\text{noise}} \right), \quad (1.11)$$

where j_t characterizes the underlying population of neural currents and comprises the number, location, strength and direction of current dipoles (i.e. $j_t = (N_t, r_t^{(1:N_t)}, q_t^{(1:N_t)})$).

Two approaches to modelling the dynamics of the underlying system of current dipoles have been considered in the literature. In both cases, the dynamics are treated as Markovian and the transition kernel is written as a weighted mixture of three components: birth in which an additional dipole becomes active, death in which an existing dipole becomes inactive and a more neutral component in which no dipoles appear or disappear. The difference between the two approaches lies essentially within this third, neutral component.

Given the interpretation of current dipoles as the activity of a fixed neural population, it seems natural to allow only the dipole moment of an existing dipole to change at any given time instant and to keep the location fixed. However, doing this introduces a substantial degree of degeneracy into the transition kernel and such degeneracy can dramatically impede the performance of SMC methods in a filtering setting. Most work on this subject has,

therefore, allowed the locations as well as the dipole moments of existing dipoles to evolve according to random walk dynamics (such approaches date back at least as far as [6]); we term this model the *random walk model*. More recent work has demonstrated that it is possible, with care, to implement particle filters for a model in which dipole locations are not permitted to move over time; we term this the *static model*.

1.5.3.2 Methodology

The approach developed in [64] makes use of a number of techniques to obtain substantially better performance than a simple particle filter. These changes are essential to obtain adequate performance with the static model. In particular, it attempts to approximate the locally optimal proposal distribution (cf. [16]), and it employs MCMC steps to improve sample diversity and to allow, to some extent, information provided by the most recent observations to influence estimates associated with the state a few time steps back (unlike a simple SIR algorithm).

In order to avoid entering into a detailed discussion of the technical issues involved in this algorithm, only a summary of the key features of the approach is provided here, noting that full details and a more complete explanation are provided in [64].

As the algorithm is performing particle filtering on a moderately high dimensional space, it is critical to employ as good a proposal distribution as possible in order to obtain importance weights which are as stable as possible (just as the quality of a simple importance-sampling estimator is determined by the quality of the proposal, the performance of a sequential Monte Carlo algorithm can be very heavily influenced by the choice of proposal distribution). In the MEG context, the underlying dynamics can be decomposed as a mixture of three components (birth, death, or steady number of dipoles) and it is natural to employ a proposal of the same form. An approach which [64] found gave a reasonable compromise between proposal quality, computational cost and implementation effort was to employ a mixture distribution. The components of this mixture being the system dynamics for the neutral component, a birth component in which the new dipole is most likely to appear at regions of the brain in which a simple Tychonoff regularized inverse takes large values and a death component which is most likely to eliminate those dipoles which make the smallest contribution to explaining the observed data.

The use of MCMC steps within SMC was first proposed by [27] who gave the approach the name “resample-move” as the original formulation depended upon only those steps. In the MEG context, the use of MCMC moves on the temporal path space allows for data about the location of dipoles to be incorporated retrospectively into estimates. By incorporating such updates into the algorithm, it is (loosely speaking) possible for it to change the estimated location of a dipole *not* by allowing that dipole to move but by updating its estimate to reflect that it was *always* at a different location from that

previously estimated. The particular construction used ensures that it remains a consistent (in the number of samples used within the SMC algorithm) estimator of the posterior density associated with the static model. Algorithm 1.4 shows the approach in outline.

Algorithm 1.4 Outline of the Resample-Move algorithm

```

Sample  $J_{0,0}^1, \dots, J_{0,0}^N \stackrel{\text{iid}}{\sim} p(j_0)$ .
for  $i = 1, \dots, N$ : Set  $W_0^i = 1/N$ .
for  $n = 1, \dots, n_{\max}$ :
  for  $i = 1, \dots, N$ :
    Set  $\tilde{J}_{n,0:n-1}^i = J_{0:n-1}^i$  and sample  $\tilde{J}_{n,n}^i$  from  $q(J_n | \tilde{J}_{n,0:n-1}^i, b_t)$ ,
    Compute  $\tilde{W}_n^i = \frac{p(b_n | \tilde{J}_{n,n}^i) p(\tilde{J}_{n,n}^i | \tilde{J}_{n,n-1}^i)}{q(\tilde{J}_{n,n}^i | \tilde{J}_{n,n-1}^i, b_n)}$ 
  end for
  for  $i = 1, \dots, N$ :  $W_n^i = \tilde{W}_n^i / \sum_j \tilde{W}_n^j$ 
  Resample  $\{\tilde{J}_{n,0:n}^i, W_n^i\}_{i=1}^N$  to obtain  $\{\bar{J}_{n,1:n}^i, 1/N\}_{i=1}^N$ .
  for  $i = 1, \dots, N$ : Sample  $J_{n,0:n}^i \sim K_n(\bar{J}_{n,0:n}^i, \cdot)$ , a  $p(j_{0:n} | b_{1:n})$ -invariant
    Metropolis-Hastings kernel.
end for

```

1.5.4 Example

This section illustrates the relative behaviour of the static and dynamic dipole models within a particle filtering framework as well as the dynamic Statistical Parametric Mapping (dSPM) method of [11]. dSPM is a well-known method in which Tychonoff regularization is applied independently at each time point, and the so-obtained estimate of the electrical current distribution is standardized by dividing by a location-dependent estimate of the noise variance.

Data from [64], which describes the data acquisition in detail:

This data comprised measures from a Somatosensory Evoked Field (SEF) mapping experiment. The recordings were performed after informed consent was obtained, and had prior approval by the local ethics committee. Data were acquired with a 306-channel MEG device (Elekta Neuromag Oy, Helsinki, Finland) comprising 204 planar gradiometers and 102 magnetometers in a helmet-shaped array. The left median nerve at wrist was electrically stimulated at the motor threshold with an interstimulus interval randomly varying between 7.0 and 9.0 s. The MEG signals were filtered to 0.1-200 Hz and sampled at 1000 Hz. Electrooculogram (EOG) was used to monitor eye movements that might produce artifacts in the MEG recordings; trials with EOG or MEG exceeding 150 mV or 3 pT/cm, respectively, were excluded and 84 clean trials were averaged. To reduce external interference, signal space separation method [67] was applied to the average. A 3D

digitizer and four head position indicator coils were employed to determine the position of the subject's head within the MEG helmet with respect to anatomical MRIs obtained with a 3-Tesla MRI device (General Electric, Milwaukee, USA).

were used to conduct this comparison. The original work features a more comprehensive comparison.

Figure 1.5 shows estimates of the activity at various points in the time series obtained using three different methods (see [64] for further details of the precise estimators employed and a wider comparison). From this sequence of

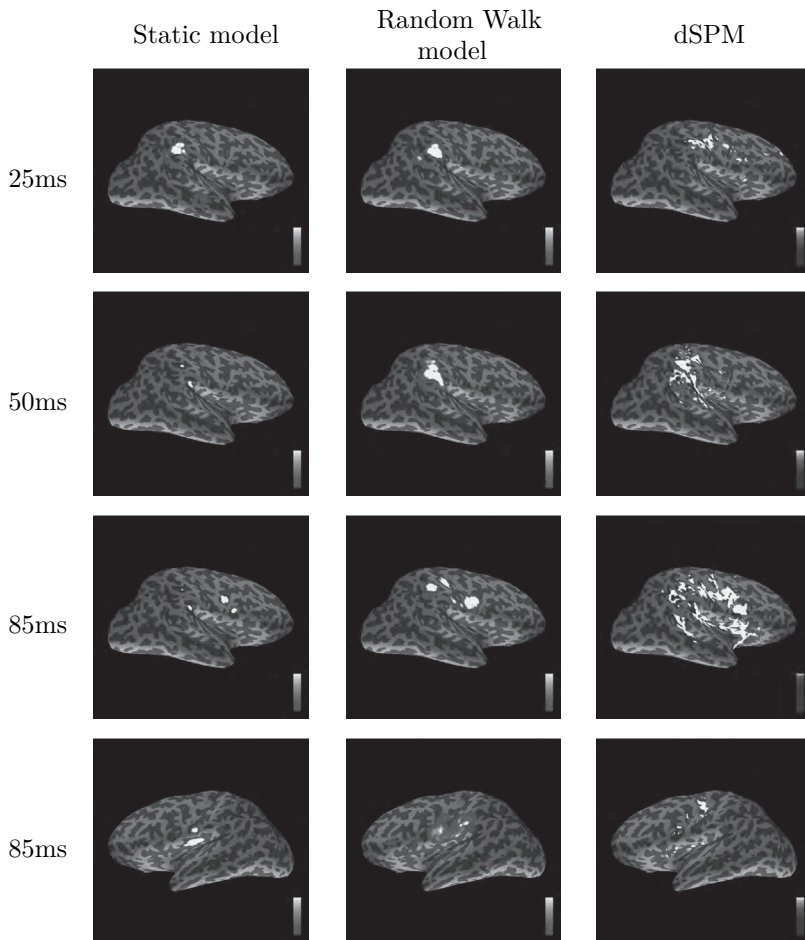


FIGURE 1.5

SEF data. Static Model (left), Random Walk Model (centre) and dSPM estimate (right). Data presented on an inflated brain representation map.

images one can discern some significant features of the methods: both methods based upon the filtering of small numbers of latent current dipoles produced more concentrated estimates of activity location and, in particular, the static model leads to more concentrated estimates than the random walk model (this is not due to the particle system becoming degenerate but to the accumulation of information over time).

It is not possible, of course, to determine how the estimated dynamics of the system behave by looking at these static images. In fact, a major advantage of the static model is that using it eliminates from the estimation non-physical artefacts which are obtained when using the random walk model. In particular, whenever one neural population ceases to be active and another begins at a similar time there are two possible explanations which are compatible with the random walk model: that one dipole has become inactive and another active at similar times *or* that the dipole has *moved* across the brain volume. Only the first of these explanations is biophysically plausible given the interpretation of these current dipoles and only this explanation is compatible with the static model. As such estimated time series obtained from the static model admit much simpler interpretations.

The ability to analyze and produce more scientifically plausible models can only lead to further understanding of the underlying mechanisms of neuronal behaviour. Future research in this area will undoubtedly involve combining complementary imaging modalities, and models such as those proposed above will be needed in order to provide understanding of possible mechanisms to allow this combination.

1.6 Conclusions

Bayesian methodology is at the forefront of neuroimaging analysis and methods, and the recent advances in computational algorithms for high dimensional data can only further enhance its role. This article has outlined only a tiny fraction of the possible applications of Bayesian techniques in Neuroscience. However, they have each demonstrated possible solutions to perceived difficulties in Bayesian modelling. It has been shown that models need not deal with stationary data with known designs, but can, by careful modelling, identify the underlying changes in the system from the data. In addition, the computational cost of Bayesian techniques does not inherently prohibit their usage in even truly massive datasets. Finally, it has been shown that models can be found that are both computationally tractable but which also allow an inherent interpretability to their results.

While in the examples here, the approaches outlined have been based on SMC, it is not the intention of this chapter to imply that this is the only computational methodology suitable for the Bayesian analysis of brain imaging

data. Indeed, almost all Bayesian computational approaches currently available have been used successfully in the analysis of neuroimaging data in different contexts. These range from using modelling frameworks which admit analytic solutions [69], to Variational Bayes approaches [20] to MCMC over the whole brain [19], with these three examples being a tiny proportion of the vast literature. Much more important than the particular choice of computation technique is that the model is suitable for the inference problem at hand. This is especially true in the current era of parallel implementations of many computational approaches [41]. However, SMC has been seen above to be a useful approach in a number of situations.

It is hoped that this article has demonstrated some of the power of Bayesian statistics to develop good solutions to real problems in the world of neuroscience and to do so with reasonable implementation effort in a timely fashion. The approaches described in this article are relatively recent and it is clear that there is substantial space for further exploration, particularly with regard to the incorporation of spatial information in the models considered (possibly in similar ways to those specified in structural MRI [19]). In addition, Bayesian methodology even in situations where the likelihood is difficult to specify is a growing area of research with direct application in Neuroscience [5]. It is our view that with the availability of fast computers and the methodology of modern computational statistics, the field of neuroimaging is amenable to substantial methodological advances over the course of the coming years and it is probable that Bayesian methods will be fundamentally involved in many of these advances.

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25 25: Empirical Bayes Methods for the Transformed Gaussian Random Field Model with Additive Measurement Errors

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26 26: Mixture Kalman Filters and Beyond

26.1 Introduction

The discrete time general state-space model is a flexible framework to deal

with the nonlinear and/or non-Gaussian time series problems. However, the

associated (Bayesian) inference problems are often intractable. Additionally,

for many applications of interest, the inference solutions are required to be

recursive over time. The particle filter (PF) is a popular class of Monte Carlo

based numerical methods to deal with such problems in real time. However,

PF is known to be computationally expensive and does not scale well with

the problem dimensions. If a part of the state space is analytically tractable

conditioned on the remaining part, the Monte Carlo based estimation is then

confined to a space of lower dimension, resulting in an estimation method

known as the Rao-Blackwellized particle filter (RBPF).

In this chapter, we present a brief review of Rao-Blackwellized particle fil

tering. Especially, we outline a set of popular conditional tractable structures

admitting such Rao-Blackwellization in practice. For some special and/or rel

atively new cases, we also provide reasonably detailed descriptions. We confine

our presentation mostly to the practitioners' point of view.

26.1.1 Problem background

Let us start with the following discrete time general state-space model relating

the latent state $x_k \in \mathbb{R}^n$ at time step k , to the observation $y_k \in \mathbb{R}^m$ as $x_k = f(x_{k-1}, w_k)$, (26.1a) $y_k = h(x_k, e_k)$, (26.1b)

where $f(x_{k-1}, w_{k-1})$ describes how the state propagates driven by the pro

cess noise w_{k-1} , and $h(x_k, e_k)$ describes how the measurements relate to the

state and how the measurement is affected by noise, e_k . Note that the la

tent state x_k in (26.1) follows a Markovian model and the observation y_k

is conditionally independent given x_k . The model (26.1) can be equivalently

characterized in terms of the known state transition density $p(x_k | x_{k-1})$ and

the observation likelihood $p(y_k | x_k)$. For this model, given the density for ini

tial state (i.e., $p(x_0)$) and a stream of observations $y_{0:k}$, $\{y_0, y_1, \dots, y_k\}$

up to time k , carrying the information about the state, one typical objective

is to optimally estimate the sequence of posterior densities $p(x_{0:k} | y_{1:k})$ and

their marginals $p(x_k | y_{1:k})$. The latter problem is generally referred to as the

filtering problem. The filtering problem appears naturally in many applica

tions of interest in applied science, econometrics, and engineering. Moreover,

for certain applications (e.g., target tracking), the above densities need to be

propagated recursively in real time. The exact posterior densities $p(x_k | y_{1:k})$

(also known as filter densities) can be obtained analytically and recursively

over time, only for a few special cases, including the linear Gaussian state where the exact posterior is given by the celebrated

Kalman filter (KF) [4, 48].

However, contrary to Kalman's linear-Gaussian assumptions (i.e., $f(\cdot)$

and $h(\cdot)$ are linear, w_k , v_k , and x_0 are Gaussian), many real world systems

are nonlinear-non-Gaussian in their nature. Optimal estimates for such models

can generally not be obtained in closed form and as a result, many approxi

mative methods have been proposed. Most of these approaches are centered

around approximating the state-space models with a linear-Gaussian form

(e.g., the extended Kalman filter, Gaussian sum filters, and nonlinear Kalman

filters based on different numerical quadrature rules). Since the early 1990's,

a class of sequential Monte Carlo simulation based filters, popularly known as

particle filters have appeared to dominate this field due to their flexibility in

adapting nonlinearity and/or non-Gaussianity without any ad hoc model as

sumptions. Particle filtering methods were introduced in their modern forms

by the seminal contributions of Gordon et al. [20]. In this

framework, the

posterior distribution is approximated by an empirical distribution formed by

(weighted) random samples, called particles, and this empirical distribution

can be propagated over time recursively (see e.g., [8, 15, 19, 21] for details).

Although the PF has been around for a while, it is computationally de

manding and notably, it has severe limitations when scaling to higher dimen

sions [14, 33, 44]. As many important practical problems are high-dimensional

in their nature, this poses a limitation on the general applicability of the PF.

However, for many practical applications, severe nonlinearity and/or non

Gaussianity often affects only a small subset of the states. The rest of the

states are often mildly nonlinear and/or non-Gaussian. Then one practical

solution to the high dimensional problems can be given by first employing a

particle filter to the highly nonlinear/non-Gaussian part of the state space, and

conditioned on this, apply an (extended) Kalman filter (or variant thereof) to

the remaining state space. This technical enabler is often known as the mixture

Kalman filter (MKF) [10].

The mixture Kalman filter requires a special model structure, i.e., , a part

of the state space is required to be conditionally

linear-Gaussian. Then one

can conditionally apply the KF, which is analytically tractable. The immedi

ate and foreseeable advantage is that solving part of the problem analytically

leaves only a part of the state vector to be solved approximately using par

ticle techniques, hence requiring fewer particles. However, this conditional

tractability is not limited to only LGSSM (as in the MKF) and it has greater

connotation in terms of efficiency improvement. When it is possible to exploit

such conditional tractability, the Monte Carlo based estimation is confined

to lower dimensional space. Consequently, the estimate obtained is often bet

ter and never worse than the estimate provided by the PF targeting the full

state-space model. This efficiency arises as an implication of the so called Rao

Blackwellization. For this reason, the resulting filtering problem is also known

as Rao-Blackwellized particle filtering [18]. in Bayesian

26.1.2 Organization

The rest of the chapter is organized as follows. In order to make the chap

ter self containing, we start with brief but necessary backgrounds on PF and

Rao-Blackwellization, in Section 26.2 and Section 26.3, respectively. This is

followed by a detailed description on the models for mixture Kalman filter in

Section 26.4. As pointed out earlier, RBPF is not only confined to the MKF

and therefore we outline other popular RBPF approaches in Section 26.5.

Next, we present a real life application in Section 26.6, where such model

frameworks are put in place. For many practical applications, where con

ditional tractability is missing, the marginalization step can be carried out

approximately using Monte Carlo integrations. This idea has led to the emer

gence of ‘approximated RBPF’, which is briefly discussed in Section 26.7.

Finally, this is followed by the concluding remarks in Section 26.8.

26.2 Particle Filtering

In PF, the posterior distribution associated with the density $p(x_{0:k} | y_{1:k})$ is

approximated by an empirical distribution represented by a set of $N (\gg 1)$

weighted particles (samples) as $P^N(dx_{0:k} | y_{1:k}) = N \sum_{i=1} \tilde{w}^{(i)}_{0:k} \delta_{x^{(i)}_{0:k}}(dx_{0:k})$, (26.2)

where $\delta_{x^{(i)}_{0:k}}(A)$ is a Dirac measure for a given $x^{(i)}_{0:k}$ and a measurable set A .

The $\tilde{w}^{(i)}_{0:k}$ is the weight associated with each particle $x^{(i)}_{0:k}$, such that $\tilde{w}^{(i)}_{0:k} \geq 0$ and $\sum_{i=1}^N \tilde{w}^{(i)}_{0:k} = 1$.

Given this representation, one can approximate the marginal

distribution associated with $p(x_k | y_{1:k})$ as $P^N(dx_k | y_{1:k}) = N \sum_{i=1} \tilde{w}^{(i)}_{0:k} \delta_{x^{(i)}_k}(dx_k)$, (26.3)

and expectations of the form $I(g_k) = \int g_k(x_{0:k}) p(x_{0:k} | y_{1:k}) dx_{0:k}$ (26.4)

as $I^N(g_k) = \int g_k(x_{0:k}) P^N(dx_{0:k} | y_{1:k}) \approx N \sum_{i=1} \tilde{w}^{(i)}_k g_k(x^{(i)}_{0:k})$. (26.5)

Although the distribution $P^N(dx_{0:k} | y_{1:k})$ does not admit a well defined density

with respect to the Lebesgue measure, the density $p(x_{0:k} | y_{1:k})$ is empirically $p^N(x_{0:k} | y_{1:k})$ given by $p^N(x_{0:k} | y_{1:k}) = N \sum_{i=1} \tilde{w}^{(i)}_k \delta(x_{0:k} - x^{(i)}_{0:k})$, (26.6)

where $\delta(\cdot)$ is the Dirac-delta function. The notation used in (26.6) is not

mathematically rigorous; however, it is intuitively easier to follow than the

more rigorous measure theoretic notations. This is especially useful when we

are not concerned with theoretical convergence studies.

Now suppose at time $(k - 1)$, we have a weighted particle approximation of the posterior $p(x_{0:k-1} | y_{1:k-1})$ as $P^N(dx_{0:k-1} | y_{1:k-1}) = \sum_{i=1}^N \tilde{w}^{(i)}_{k-1} \delta(x_{0:k-1} - x^{(i)}_{0:k-1})$. With the arrival of a new measurement y_k , we

want to approximate $p(x_{0:k} | y_{1:k})$. This is achieved by propagating the particles in time using importance sampling and resampling steps. First, the particles are propagated to time k by sampling a new state $x^{(i)}_k$ from a proposal kernel $\pi(x_k | x^{(i)}_{0:k-1}, y_{1:k})$ and then setting $x^{(i)}_{0:k} = (x^{(i)}_{0:k-1}, x^{(i)}_k)$. Since we have

the posterior recursion as

$p(x_{0:k} | y_{1:k}) \propto p(y_k | x_{0:k}, y_{1:k-1}) p(x_k | x_{0:k-1}, y_{1:k-1}) p(x_{0:k-1} | y_{1:k-1})$ (26.7)

and using (26.1), the corresponding (normalized) weights of the particles are

$\tilde{w}^{(i)}_k = \frac{p(y_k | x^{(i)}_{0:k}, y_{1:k-1}) p(x^{(i)}_k | x^{(i)}_{0:k-1}, y_{1:k-1}) \tilde{w}^{(i)}_{k-1}}{\sum_{i=1}^N p(y_k | x^{(i)}_{0:k}, y_{1:k-1}) p(x^{(i)}_k | x^{(i)}_{0:k-1}, y_{1:k-1}) \tilde{w}^{(i)}_{k-1}}$

updated as $w^{(i)}_k \propto \tilde{w}^{(i)}_{k-1} p(y_k | x^{(i)}_k) p(x^{(i)}_k | x^{(i)}_{k-1}) \pi(x^{(i)}_k | x^{(i)}_{0:k-1}, y_{1:k})$; $\tilde{w}^{(i)}_k = w^{(i)}_k / \sum_{j=1}^N w^{(j)}_k$. (26.8)

Now, to avoid carrying trajectories with small weights, and to focus on the

ones with large weights, the particles are resampled (with replacement), when

ever necessary (see, e.g., [17] for details). The resampled particles representing

the same posterior $p(x_{0:k} | y_{1:k})$ are equally weighted. The whole procedure is

repeated sequentially over time. For a more general introduction to PF, refer

to, e.g., [19].

26.3 Rao-Blackwellization

The foundation of the Rao-Blackwellization technique used in many estima

tion and filtering applications dates back to the 1940's. In the 1940's Rao [35]

and Blackwell [7] showed that an estimator can be improved by making use of

conditional expectations, if such structure exist in the problem formulation.

They further showed how an estimator based on this knowledge can be con

structed as an expected value of a conditioned estimator. This practice has

over time become known as Rao-Blackwellization. In Bayesian theorem is often used to show how the knowledge of

a conditional expected value can improve the variance of an estimate, but

this is only a special case of a more general theorem which shows that any

convex cost-function is improved using this technique. For completeness, the

Rao-Blackwell theorem is given in Theorem 26.3.1, [28, p. 51].

Theorem 26.3.1 (The Rao-Blackwell theorem [28]). Let x be a random ob

servable with the distribution $P, \theta \in \Theta = \{P, \theta' \in \Omega\}$, and let t be sufficient

statistics for P . Let $\hat{\theta}$ be an estimator of an estimand $g(\theta)$, and let the loss

function $L(\theta, d)$ be a strictly convex function in d . Then if $\hat{\theta}$ has finite expec

tation and risk, $R(\theta, \hat{\theta}) = E(L(\theta, \hat{\theta}(x)))$ and if $\hat{\theta}_{rb}(t) = E(\hat{\theta}(x)|t)$

the risk of the estimator $\hat{\theta}_{rb}(t)$ satisfies $R(\theta, \hat{\theta}_{rb}) \leq R(\theta, \hat{\theta})$,

unless $\hat{\theta}(x) = \hat{\theta}_{rb}(t)$ with probability 1, where $A \leq B$ denotes that $B - A$ is a

positive definite matrix.

Proof. See [28].

It is worth pointing out once more that the Rao-Blackwell theorem does

not only predict a reduced risk, it also shows how to construct the estimator

to obtain the gain as an expected value. This makes the theorem very useful,

as will be exemplified a number of times in the remainder of this chapter.

An important and often used special case of Theorem 26.3.1 describes the

variance properties of a Rao-Blackwell estimator.

Corollary 1. Let $\hat{\theta}$ be an unbiased estimator that fulfills the requirements in

Theorem 26.3.1 and $\hat{\theta}^{rb}$ the Rao-Blackwell estimator, then $C(\hat{\theta}^{rb}) \leq C(\hat{\theta})$

unless $\hat{\theta}(x) = \hat{\theta}^{rb}(t)$ with probability 1.

Proof. To prove the corollary, assume the cost function $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})(\theta - \hat{\theta})^T$,

which is convex in both θ and $\hat{\theta}$, and apply Theorem 26.3.1. Finally note that

$C(\theta) = R(\theta, \hat{\theta}) = E(L(\theta, \hat{\theta}))$, which concludes the proof.

1 The statistic t is sufficient for θ if the conditional distribution of x given t does not

depend on θ . gain using Rao-Blackwellization is given by the separation

of covariance, [24], which states $C(x) = C(E(x|y)) + E(C(x|y))$,

where $C(x|y)$ denotes the covariance matrix of x conditioned on y . Assuming

that x is the state estimate, then $E(x|y)$ is the improved estimate in the

Rao-Blackwell theorem. Hence, compared to the Rao-Blackwellized estimate,

the regular estimate has an additional covariance term, $E(C(x|y))$, that by

definition is positive semi-definite. The covariance gain obtained with a Rao

Blackwellized estimator comes from eliminating this term in the covariance

expression.

26.4 Models for Mixture Kalman Filters

The mixture Kalman filter was mentioned earlier as an enabling technique

for PF to reasonably high dimensional problems. Since MKF

uses conditional

KF, it essentially requires a conditionally linear Gaussian state-space model

(CLGSSM). This is an important class of models with many practical applica

tions which has received extensive research attentions in the past [3, 21, 22, 29].

In the sequel, we outline two different model structures conforming to this

class: the hierarchical conditionally linear Gaussian state-space model and

the mixed linear/nonlinear Gaussian state-space model.

26.4.1 Hierarchical conditionally linear Gaussian model

The hierarchical conditionally linear Gaussian model (HCLGM) is, as the

name suggests, a hierarchical model structure in two layers where a latent

process c_k propagates according to a Markov kernel, independent of the second

part of the latent variable x_k . Conditioned on c_k , (x_k, y_k) follows a linear

Gaussian state-space model, parametrized in c_k . Mathematically the HCLGM

can then be described by $c_{k+1} \sim p(c_{k+1} | c_k)$ (26.9a) $x_{k+1} = f(c_k) + F(c_k)x_k + G(c_k)w_k$ (26.9b) $y_k = h(c_k) + H(c_k)x_k + H_e(c_k)e_k$, (26.9c)

where $p(c_{k+1} | c_k)$ indicates that c_k is propagated using a Markov kernel; w_k

and e_k are Gaussian process and measurement noises, respectively. A typical

example of an HCLGM is a linear system with a scaled mixture of Gaussian

noises (e.g., Student's t noise distribution [30, 41]);

here conditioned on $c_{0:k}$,

the noise can be suitably represented as Gaussian and consequently KF can be

applied on the conditionally linear-Gaussian part. A graphical illustration of in Bayesian

FIGURE 26.1

Graphical representation of dependencies in hierarchical conditionally linear

Gaussian model (HCLGM) (26.9) and the mixed linear/nonlinear Gaussian

state-space model (ML/NGSSM) (26.11).

the hierarchical model structure is given in Figure 26.1(a). This class of models

is also known as the conditionally dynamic linear model (CDLM) [10]. For

the inference task, we can decompose the joint density $p(c_{0:k}, x_k | y_{1:k})$, $p(c_{0:k}, x_k | y_{1:k}) = p(c_{0:k} | y_{1:k}) p(x_k | c_{0:k}, y_{1:k})$, (26.10)

where we target the propagation of $p(c_{0:k} | y_{1:k})$ using a PF and for each se

quence $c_{0:k}$, the propagation of $p(x_k | c_{0:k}, y_{1:k})$ using a KF.

26.4.2 Mixed linear/nonlinear Gaussian state-space model

Another common model following a CLGSSM structure is the mixed

linear/nonlinear Gaussian state-space model (ML/NGSSM) [43]. The

ML/NGSSM is a state-space model where it is possible to divide the state

vector into two parts, $x_k = (x_{n,k} \times x_{l,k})$, such that conditioned on $x_{n,k}$, $x_{l,k}$ assumes a

linear-Gaussian state-space model, $x_{n,k+1} = f_n(x_{n,k}) + F_n(x_{n,k})x_{l,k} + w_{n,k}$ (26.11a) $x_{l,k+1} = f_l(x_{n,k}) +$

$$F_1(x_{n:k})x_{l:k} + w_{l:k} \quad (26.11b) \quad y_k = h(x_{n:k}) + H(x_{n:k})x_{l:k} + e_k, \quad (26.11c)$$

where $w_{n:k}$, $w_{l:k}$, and e_k are mutually independent Gaussian noise processes with

variance $Q_{n:k}$, $Q_{l:k}$, and R_k , respectively. Furthermore, $x_{l:0}$ is assumed to follow a

Gaussian distribution. The distribution of $x_{n:0}$ can be arbitrary, but is assumed

2 Although we restrict c_k to be a continuous variable for CDLM, in [10] CDLM is defined

in a broader sense such that c_k can be either a continuous or discrete variable. The above

model corresponding to a discrete c_k is popularly known as switching linear Gaussian system

(or jump Markov linear system), which, according to our notations, is a special case of model

(26.5.2), discussed later. a model is referred as the partial conditionally dynamic

linear model (partial CDLM). However, [10] did not consider the influence

of $x_{l:k}$ in the evolution of $x_{n:k}$. This influence was first considered in [31] and

was further popularized in [43]. This framework is extended to the dependent

noise processes in [37]. The interconnection between the two parts of the state

space, $x_{n:k}$ and $x_{l:k}$ is best illustrated in a graphical model as in Figure 26.1(b).

The nonlinear filtering problem can be decomposed into two sub-problems, $p(x_{l:k}, x_{n:0:k} | y_{1:k}) = p(x_{n:0:k} | y_{1:k}) p(x_{l:k} | x_{n:0:k}, y_{1:k})$, (26.12)

where the propagation of $p(x_{n:0:k} | y_{1:k})$ is targeted using a PF and conditioned on

each sequence $x_{n:0:k}$, $p(x_{l:k} | x_{n:0:k}, y_{1:k})$ can be

propagated analytically using KF.

The algorithm descriptions for this model are rather technical. The mixture

Kalman filter here, when a bit simplified, works in the following way: The

nonlinear part of the problem is solved using a particle filter. Attached to

each particle is a KF, which estimates the (conditionally) linear part of the

state. To compensate for the interactions between the linear and nonlinear

parts of the state vector, a virtual measurement update must be applied to

the KF each time the particle filter is propagated forward in time.

Later [25, 26] reformulated this RBPF scheme in terms of a stochastic

bank of KFs, which highlighted the structural components of the solution.

In essence, this RBPF scheme becomes a Kalman filter bank with stochas

tic pruning and weight computations provided by the PF. The algorithmic

summary of this approach is given in Algorithm 26.1.

The ML/NGSSM as in (26.11) is quite common, especially in tracking and

navigation applications, where the unknown state vector typically consists

of position, velocity, and sometimes acceleration. When fitted into such a

framework, the position typically ends up in the nonlinear part of the state

space, whereas the velocity and acceleration end up in the

linear part. These

applications were considered in [23, 31] under the name
marginalized particle

filter. 3

This model structure is also extensively used in robotics
to address the

simultaneous localization and mapping (SLAM) problem. Here
one attempts

to localize oneself in an unknown environment and at the
same time, build

a map of the surrounding environment. In essence, this
yields a state-space

model with a state comprised of the kinematic state of
oneself, and the map,

which grows over time. Fortunately, conditioned on the
state, the map fits in

a linear-Gaussian model which allows for
Rao-Blackwellization. This fact is

utilized in the FastSLAM algorithm which is an MKF in
disguise [29].

3 The reason behind that name is that the linear (or, more
generally, analytical) part of

the state vector is ‘marginalized’ (or, integrated out) in
closed form to obtain the posterior

distribution of the nonlinear (or, non-analytical) part. in
Bayesian

Algorithm 26.1 Rao-Blackwellized PF [26]

For the system (26.11).

Initialization: For $i = 1, \dots, N$, let $x_{p(i)}|_{-1} \sim \pi$
 n_0 , $x_{k(i)}|_{-1} = x_k|_0$, $P_{k(i)}|_{-1} = P_1|_0$, $w_0|_{-1} =$
 $1/N$.

Iterations: For $k = 0, \dots$ do

- Measurement update, for each particle $i = 1, \dots, N$ do $\omega(i)k|k \propto N(y_k; \hat{y}(i)k, S(i)k) \cdot \omega(i)k|k-1 \times$
 $x(i)k|k = x(i)k|k-1 + K(i)k(y_k - \hat{y}(i)k)P(i)k|k$
 $= P(i)k|k-1 - K(i)kS(i)kK(i)k^T$,

where $\hat{y}(i)k = h(x_n(i)k|k-1) + H(x_n(i)k|k-1)x_l(i)k|k-1$
 $S(i)k = H(x_n(i)k|k-1)P(i)k|k-1H^T(x_n(i)k|k-1) + R_k$
 $K(i)k = P_l(i)k|k-1H^T(x_l(i)k|k-1)(S(i)k)^{-1}$.

- If necessary, resample the particles using standard techniques. When se

lecting a particle $x_n(i)k|k$, select the matching linear states $x_l(i)k|k$ and $P_l(i)k|k$,

and set $\omega(i)k|k$ appropriately.

- Time update, for each particle $i = 1, \dots, N$ do $x_n(i)k+1|k = f_n(x_nk|k) + F_n(x_n(i)k|k)x_l(i)k|k$
 $x_l(i)k+1|k = f_l(x_nk|k) + F_l(x_n(i)k|k)x_l(i)k|k$
 $P_n(i)k+1|k = F_n(x_n(i)k|k)P_l(i)k+1|k(F_n(x_n(i)k|k))^T + Q_n$
 $kP_n(i)k+1|k = F_n(x_n(i)k|k)P_l(i)k+1|k(F_l(x_n(i)k|k))^T + Q_l$
 k .

With $\xi(i)k+1 \sim N(0, P_n(i)k+1|k)$ do $x_n(i)k+1|k = x_n(i)k+1|k + \xi(i)k+1$
 $x_l(i)k+1|k = x_l(i)k+1|k + (P_n(i)k+1|k)^T(P_n(i)k+1|k)^{-1}\xi(i)k+1$
 $P_l(i)k+1|k = P_l(i)k+1|k - (P_n(i)k+1|k)^T(P_n(i)k+1|k)^{-1}P_n(i)k+1|k$.

26.5 Beyond Mixture Kalman Filter in Practice

Note that the key to efficient Rao-Blackwellization is a (conditionally)

tractable substructure, which is not necessarily limited to the conditionally

linear-Gaussian state-space model. In this section, we present some other ex

isting Rao-Blackwellization approaches that are popular in practice.

26.5.1 Hierarchical conditionally finite HMM

Similar to the LGSS model, when the latent state evolves according to a finite

state Markov chain, the posterior of the state given the observations, can

be obtained in closed form using the hidden Markov model (HMM) forward

algorithm [34]. This model structure is known as the finite state-space HMM.

Now if part of the state space follows such finite state-space HMM, it is then

possible to exploit an RBPF formulation.

One such RBPF framework appears when the state space follows a finite

state HMM, whose parameters are modulated by another Markovian process

[18]. This model structure is denoted the hierarchical conditionally finite HMM

(HCfHMM) and can be described as $p(c_k | c_{k-1})$, $p(r_k | c_k, r_{k-1})$, $p(y_k | c_k, r_k)$, (26.13)

where r_k is a finite state-space Markov chain, whose parameter at time k

depends on a Markov process c_k . Both r_k and c_k are latent and are related

to the observation y_k . Note that c_k is at the top of the hierarchy and the

evolution of c_k is independent of r_k . The graphical representation is shown in

Figure 26.2(a). Now conditioned on c_k , r_k follows a finite state-space Markov

chain of known parameters. Therefore one can implement a Rao-Blackwellized

particle filter, where the PF targets only $p(c_{0:k} | y_{0:k})$ and where $p(r_k | c_{0:k}, y_{0:k})$

is analytically tractable.

FIGURE 26.2

Graphical representation of the hierarchical conditionally finite HMM

(HCfHMM) and multiple switching nonlinear dynamics model (MSNDM). in Bayesian

26.5.2 Multiple switching nonlinear dynamic models

In many practical applications of engineering and econometrics, one often

deals with nonlinear dynamic systems involving both a continuous value tar

get state and a discrete value regime variable. Such descriptions imply that

the system can switch between different nonlinear dynamic regimes, where

the parameters of each regime is modulated by the corresponding regime vari

able. The regime variable also evolves dynamically according to a finite state

Markov chain. Both the target state and regime variable are latent and are re

lated to the noisy observations. This type of model is referred to as (Markov)

regime switching nonlinear dynamic models or hybrid nonlinear state-space

models.

The probabilistic description of the above model can be given by $\Pi(r_k | r_{k-1})$, $p_{\theta}^{r_k}(x_k | x_{k-1}, r_k)$, $p_{\theta}^{r_k}(y_k | x_k, r_k)$, (26.14)

where $r_k \in S$, $\{1, 2, \dots, s\}$, is a (discrete) regime indicator variable with fi

nite number of regimes (i.e., , categorical variable), $x_k \in \mathbb{R}^n$ is the (con

tinuous) state variable. As the system can switch between

different dynamic

regimes, for a given regime variable $l \in S$, the corresponding dynamic regime

can be characterized by a set of parameters θ_l . Both x_k and r_k are latent vari

ables, which are related to the measurement $y_k \in \mathbb{R}^{n_y}$. The evolution of the

regime variable r_k is commonly modeled by a homogeneous (time invariant)

first order Markov chain with transition probability matrix (TPM) $\Pi = [\pi_{ij}]$ $i, j \in S$, $P(r_k = j | r_{k-1} = i)$ $(i, j \in S)$, (26.15a) $\pi_{ij} \geq 0$; $\sum_{j=1}^S \pi_{ij} = 1$. (26.15b)

This model is represented graphically in Figure 26.2(b). Note that the above

model can easily be generalized to the cases involving time dependent dynam

ics and exogenous inputs. For the model, given the densities for the initial

state $\{r_0, x_0\}$ and the measurements up to a time k , our interest lies in es

timating $P(r_k | y_{1:k})$ and $p(x_k | y_{1:k})$ recursively. However, the above posteriors

are computationally intractable. In connection to this, we note from (26.14)

that conditioned on the sequence $x_{0:k}$, r_k follows a finite state hidden Markov

model. As a result, $P(r_k | x_{0:k}, y_{1:k})$ is analytically tractable [36]. Using this an

alytical substructure, it is possible to implement an efficient RBPF scheme 4.

We provide below a brief description of RBPF implementation for the same.

26.5.2.1 An RBPF implementation

The initial densities for the state and the regime variables are given by $p(x_0)$

and $P(r_0)$, $P(r_0 | x_0)$, respectively, which can be arbitrary but is assumed to

be known. We further assume favorable mixing conditions as in [13].

4 It should be noted that the switching linear Gaussian system is a special case of this

model structure, where the inference problem is popularly addressed using an alternative

RBPF scheme: the propagation of $P(r_{0:k} | y_{1:k})$ is targeted by a PF and conditioned on $r_{0:k}$,

$p(x_k | r_{0:k}, y_{1:k})$ is obtained analytically using KF methods at time $(k - 1)$. We consider the extended target

density $p(r_{k-1}, x_{0:k-1} | y_{1:k-1})$ which can be decomposed as

$$p(r_{k-1}, x_{0:k-1} | y_{1:k-1}) = p(r_{k-1} | x_{0:k-1}, y_{1:k-1}) p(x_{0:k-1} | y_{1:k-1}) \quad (26.16)$$

The posterior propagation of the latent state x_{k-1} can be targeted through

a PF, where $p(x_{0:k-1} | y_{1:k-1})$ is represented by a set of N weighted random

particles as $p(x_{0:k-1} | y_{1:k-1}) \approx N \sum_{i=1} w^{(i)}_{k-1} \delta(x_{0:k-1} - x^{(i)}_{0:k-1})$. (26.17)

Now conditioned on $\{x_{0:k-1}, y_{1:k-1}\}$, the regime variable r_{k-1} follows a fi

nite state-space HMM. Consequently, $p(r_{k-1} | x_{0:k-1}, y_{1:k-1})$ is analytically

tractable 5, which is represented as $q^{(i)}_{k-1|k-1}(l)$, $P(r_{k-1} = l | x^{(i)}_{0:k-1}, y_{1:k-1})$, (26.18)

for $l \in S$ and $i = 1, \dots, N$. Now using (26.17) and (26.18), the extended target

density in (26.16) can be represented as $[x^{(i)}_{0:k-1}, w$

$$q(i)_{k-1}, \{q(i)_{k-1}|k-1(l)\}_{l=1}^L, N, i=1, \dots, N. \quad (26.19)$$

Now having observed y_k , we want to propagate the extended target density

in (26.16) to time k . This can be achieved in the following steps (a)-(d):

(a) Prediction step for conditional HMM filter: This is obtained as

$$q(i)_{k|k-1}(l), P(r_k = l | x(i)_{0:k-1}, y_{1:k-1}) = \sum_{j=1}^L \pi_{jl} q(i)_{k-1|k-1}(j), (l, j) \in S. \quad (26.20)$$

(b) Prediction step for particle filter: At this stage, generate N new

samples $x(i)_k$ from an appropriate proposal kernel as $x(i)_k \sim \pi(x_k | x(i)_{0:k-1}, y_k)$.

Then set $x(i)_{0:k} = \{x(i)_{0:k-1}, x(i)_k\}$, for $i = 1, \dots, N$, representing the particle

trajectories up to time k .

(c) Update step for conditional HMM filter: Noting that

$$P(r_k = l | x_{0:k}, y_{1:k}) \propto p(y_k, x_k | r_k = l, x_{0:k-1}, y_{1:k-1}) \times P(r_k = l | x_{0:k-1}, y_{1:k-1}), \quad (26.21)$$

5 The 'favorable' mixing property ensures that $p(x_{0:k-1} | y_{1:k-1})$ can be well approx

imated by $p(x_{k-L:k-1} | y_{1:k-1})$, for some lag L .

Consequently, $p(r_{k-1} | x_{0:k-1}, y_{1:k-1}) \approx$

$$p(r_{k-1} | x_{k-L:k-1}, y_{1:k-1}). \text{ in Bayesian } q(i)_{k|k}(l) \propto p(y_k, x(i)_k | r_k = l, x(i)_{0:k-1}, y_{1:k-1}) q(i)_{k|k-1}(l) \quad (26.22a) \propto p_{\theta_l}(y_k | x(i)_k, r_k = l) p_{\theta_l}(x(i)_k | x(i)_{k-1}, r_k = l) q(i)_{k|k-1}(l). \quad (26.22b)$$

Now defining $\alpha(i)_{k}(l), p_{\theta_l}(y_k | x(i)_k, r_k = l) p_{\theta_l}(x(i)_k | x(i)_{k-1}, r_k = l) q(i)_{k|k-1}(l)$ (26.23)

we obtain $q(i)_{k|k}(l) = \alpha(i)_{k}(l) / \sum_{j=1}^L \alpha(i)_{k}(j)$, (26.24)

for $l \in S$ and $i = 1, \dots, N$.

(d) Update step for particle filter: As the continuous state can be recur

sively propagated using the following relation: $p(x_{0:k} | y_{1:k}) \propto p(y_k, x_k | x_{0:k-1}, y_{1:k-1}) p(x_{0:k-1} | y_{1:k-1})$, (26.25)

the corresponding weight update equation for the particle filtering is given by

$$w^{(i)}_k = p(x^{(i)}_k, y_k | x^{(i)}_{0:k-1}, y_{1:k-1}) \pi_k(x^{(i)}_k | x^{(i)}_{0:k-1}, y_{1:k}) \tilde{w}^{(i)}_{k-1}; \tilde{w}^{(i)}_k = w^{(i)}_k / \sum_{j=1}^N w^{(j)}_k, \quad (26.26)$$

where $\{\tilde{w}^{(i)}_k\}_{i=1}^N$ are the normalized weights. The numerator in (26.26) can be

obtained as

$$p(x^{(i)}_k, y_k | x^{(i)}_{0:k-1}, y_{1:k-1}) = \sum_{l=1}^s p(y_k, x^{(i)}_k | r_k = l, x^{(i)}_{0:k-1}, y_{1:k-1}) P(r_k = l | x^{(i)}_{0:k-1}, y_{1:k-1}), \quad (26.27)$$

which is basically given by the normalizing constant of (26.24). Note that

the marginal density $p(x_k | y_{1:k})$ can be obtained in the particle cloud form as

$p(x_k | y_{1:k}) \approx \sum_{i=1}^N \tilde{w}^{(i)}_k \delta(x_k - x^{(i)}_k)$. The posterior probability of the regime

variable can now be obtained as

$$P(r_k = l | y_{0:k}) = \int P(r_k = l | x_{0:k}, y_{0:k}) p(x_{0:k} | y_{0:k}) dx_{0:k} \approx N \sum_{i=1}^N q^{(i)}_k |k(l) \tilde{w}^{(i)}_k. \quad (26.28)$$

For further details and algorithmic summary, please refer to [38, 39].

26.5.3 Partially observed Gaussian state-space models

Consider the following partially observed Gaussian state-space models

$$(POGSSM) [2] \quad x_k = F_k x_{k-1} + G_k v_k, \quad (26.29a) \quad y_k = H_k$$

$$x_k + H_e k e_k, \quad (26.29b) \quad z_k \sim p(z_k | y_k), \quad (26.29c)$$

FIGURE 26.3

Graphical representation of a partially observed Gaussian state-space model

(POGSSM) and a partially observed finite HMM (POfHMM).

where x_k is a latent (Markov) state process with initial density given by a

Gaussian with mean x^0 and variance P^0 as $p(x^0) = N(x^0 | x^0, P^0)$; y_k and z_k are

latent data process and observed data process, respectively. The processes x_k

and y_k together define a standard linear Gaussian state-space model. Compared to standard LGSS model, however, one observes z_k

instead of y_k . z_k

is typically a quantized or censored version of y_k . For example, in a dynamic

probit model, we have $z_k = \{ 1, y_k > 0; 0, y_k \leq 0 \}$, (26.30)

whereas for a dynamic tobit model, $z_k = \max(y_k, 0)$. (26.31)

The graphical representation of the model is shown in Figure 26.3(a).

For inference purpose, the joint density can be decomposed as $p(x_{0:k}, y_{0:k} | z_{0:k}) = p(x_{0:k} | y_{0:k}, z_{0:k}) p(y_{0:k} | z_{0:k})$. (26.32)

Then the density $p(y_{0:k} | z_{0:k})$ can be targeted using a PF in the form of N

weighted particles as $p(y_{0:k} | z_{0:k}) \approx \sum_{i=1}^N \omega^{(i)}_k \delta(y_{0:k} - y^{(i)}_{0:k})$. Now for each sequence $y^{(i)}_{0:k}$, the conditional density $p(x_{0:k} | y^{(i)}_{0:k}, z_{0:k})$ is analytically tractable

using a KF. Finally, $p(x_{0:k} | z_{0:k})$ can be obtained by marginalizing over $y_{0:k}$ as

$$p(x_{0:k} | z_{0:k}) = \int p(x_{0:k} | y_{0:k}, z_{0:k}) p(y_{0:k} | z_{0:k}) dy_{0:k} \approx \sum_{i=1}^N \omega^{(i)}_k p(x_{0:k} | y^{(i)}_{0:k}, z_{0:k}).$$

(26.33)

Thus $p(x_{0:k} | z_{0:k})$ is obtained as random weighted mixture of Gaussians. For

algorithmic details, model extensions and applications, please refer to the

original article [2]. in Bayesian

26.5.4 Partially observed finite HMM

Consider the following model described probabilistically as $r_k \sim \pi(r_k | r_{k-1})$, $y_k \sim p(y_k | r_k)$, $z_k \sim p(z_k | y_k)$, (26.34)

where the dynamics of r_k is given by a homogeneous first order finite state

Markov chain. Here r_k and y_k together describe a partially observed finite

HMM (POfHMM). However instead of y_k , z_k is actually observed. The graph

ical representation of the model is shown in Figure 26.3(b).

Similar to (26.5.3), the joint density can be decomposed as $p(r_k, y_{0:k} | z_{0:k}) = p(r_k | y_{0:k}, z_{0:k}) p(y_{0:k} | z_{0:k})$. (26.35)

Again, the density $p(y_{0:k} | z_{0:k})$ can be targeted using a PF in the form of N

weighted particles as $p(y_{0:k} | z_{0:k}) \approx \sum_{i=1}^N \omega^{(i)}_k \delta(y_{0:k} - y^{(i)}_{0:k})$. Then for each

sequence $y_{0:k}$, $p(r_k | y_{0:k}, z_{0:k})$ can be obtained analytically using the property

of finite HMM. For details, see [2].

26.5.5 Conditionally conjugate latent process model

The key enabler here is the concept of Bayesian conjugacy. For a latent vari

able with known observation likelihood, if we can choose a suitable prior (rep

resented with a parametric distribution) such that the prior and posterior

belong to the same parametric family, then it is said to be conjugate to the

observation likelihood. As a result, the posterior can be obtained in closed

form by updating the parameters of the prior distributions. Such conjugate

priors are known to exist for the members of the exponential family of dis

tributions [6]. Using this conjugacy, one can define a latent process in such a

way that both the filtering and prediction distributions are conjugate to the

observation distribution in each time step. The latent process so defined is

called the conjugate latent process [47].

Now if the state space can be decomposed such that part of the state space

follows a conjugate latent process conditioned on the other part, one can then

marginalize this part analytically. In the particle filtering framework, this

leads to another Rao-Blackwellization approach. One typical application of

the resulting RBPF is a noise adaptive particle filtering for a general state

space model [32, 37, 42]. A brief description of the general approach is given

below.

26.5.5.1 Rao-Blackwellized noise adaptive particle filter

For many practical applications, complete knowledge of the noise processes

in the state-space models is unavailable. Typically for such cases, noises are

assumed to be generated from a known parametric family of distributions

with unspecified parameters. In such settings, one needs to estimate the noise

parameters along with the latent state using the available noisy observations. as noise adaptive filtering. Moreover, the noise may

be non-stationary or state dependent, which prevents off-line tuning. For the

above reasons, one needs a sequential method to learn the noise parameters

using streamed sensor data.

When the noise distribution is a member of the exponential family, we can

model the unknown noise parameter as conjugate latent process. To describe

the resulting filtering problem, we first start with a description of the model

(including the noise model), which is followed by the problem definition and

implementation details.

Model description: Consider the following general state-space model relat

ing a (Markovian) latent state x_k to the observation y_k at time step k as $x_k = f_k(x_{k-1}, u_{k-1}) + g_k(x_{k-1}, u_{k-1})v_k$, (26.36a) $y_k = h_k(x_k, u_k) + d_k(x_k, u_k)w_k$. (26.36b)

Here, $f_k(\cdot)$, $h_k(\cdot)$, $d_k(\cdot)$ and $g_k(\cdot)$ are nonlinear functions of the state vector

x_k and the deterministic input u_k at time k . Here $d_k(\cdot)$ is assumed to be

invertible whereas $g_k(\cdot)$ is not, as most motion models in practice, including

those with integrators, lead to non-invertible $g_k(\cdot)$.

Noise model: We define the noise vector e_k , $[v_k^T, w_k^T]^T$ as a realization from

a distribution belonging to the exponential family as $e_k \sim p(e_k | \theta_k) = p(e_k) \exp(\eta(\theta_k) \cdot \tau(e_k) - \phi(\theta_k))$, (26.37)

where θ_k is the vector of unknown parameters, $\eta(\theta_k)$ and $\phi(\theta_k)$ are vector and

scalar valued functions of the parameters, respectively; $p(e_k)$ and $\tau(e_k)$ are

scalar and vector valued functions of the realization e_k ; and the symbol ' $\cdot$ '

denotes scalar product of two vectors. Further, following [32], assume that the

unknown noise parameters θ_k are slowly varying in time. This slowly varying

nature can arise, e.g., due to model misspecification [48].

Now for a given realization of the noise e_k , (26.37) provides the likelihood

function for the unknown noise parameter θ_k . Accordingly, we can select a

conjugate prior on θ_k as

$p(\theta_k | e_{1:k-1}) = \gamma^{-1}(V_{k|k-1}, \nu_{k|k-1}) \exp(\eta(\theta_k) V_{k|k-1} - \nu_{k|k-1} \phi(\theta_k))$, (26.38)

where $V_{k|k-1}$ is a vector of sufficient statistics and $\nu_{k|k-1}$ is a scalar counter

of the effective number of samples in the statistics. The normalization factor

$\gamma(V_{k|k-1}, \nu_{k|k-1})$ is uniquely determined by the statistics $V_{k|k-1}$ and $\nu_{k|k-1}$.

Then, the posterior density $p(\theta_k | e_{1:k})$ is also of the

form (26.38) with statistics $V_k|k = V_k|k-1 + \tau(e_k)$, $\nu_k|k = \nu_k|k-1 + 1$. (26.39)

This is a convenient result for recursive evaluation of sufficient statistics start

ing from a prior defined by V_0 and ν_0 . Further, the predictive distribution of θ_k in Bayesian

$$p(e_k | e_{1:k-1}) = \int p(e_k | \theta_k) p(\theta_k | e_{1:k-1}) d\theta_k = \gamma(V_k|k, \nu_k|k) / \gamma(V_k|k-1, \nu_k|k-1) p(e_k). \quad (26.40)$$

Now as θ_k is assumed to be slowly varying in time, to complete the state-space

model, an evolution model for θ_k (i.e., $p(\theta_k | \theta_{k-1})$) is required. Given this

evolution model, the predictive density of the parameter θ_{k+1} is obtained as $p(\theta_{k+1} | e_{1:k}) = \int p(\theta_{k+1} | \theta_k) p(\theta_k | e_{1:k}) d\theta_k$. (26.41)

If this predictive density conforms to the specification (26.38), then the noise

parameter vector follows a conjugate latent process. However, typically the

transition model $p(\theta_{k+1} | \theta_k)$ is unknown. In [32], the evolution is defined implicitly

as a Kullback-Leibler norm constraint on the time variability which leads

to an exponential forgetting mechanism operating on the sufficient statistics.

This keeps the predictive density $p(\theta_{k+1} | e_{1:k})$ in the same class as in (26.38)

with statistics updated from (26.39) as $\nu_{k+1}|k = \lambda \nu_k|k + (1-\lambda)\nu_u$, (26.42a) $V_{k+1}|k = \lambda V_k|k + (1-\lambda)V_u$, (26.42b)

where $\lambda \in (0, 1]$ is a forgetting factor and the invariant measure is assumed

to be also in the exponential form (26.38) with statistics ν_u , V_u . The details

can be found in [32].

Problem definition: We are concerned with the evaluation of the joint pos

terior density $p(x_k, \theta_k | y_{1:k})$ sequentially over time. Suppose at time $(k-1)$ we

have the posterior $p(x_{k-1}, \theta_{k-1} | y_{1:k-1})$. With the arrival of a new observation

y_k , we want to update the posterior recursively to $p(x_k, \theta_k | y_{1:k})$.

Joint estimation of state and noise parameters: The joint posterior

density at time k can be decomposed as follows: $p(x_{0:k}, \theta_{0:k} | y_{0:k}) = p(\theta_{0:k} | x_{0:k}, y_{0:k}) p(x_{0:k} | y_{0:k})$.
(26.43)

The posterior propagation of the latent state x_k is targeted through a

PF as $p(x_{0:k} | y_{0:k}) \approx \sum_{i=1}^n \omega^{(i)}_k \delta(x_{0:k} - x^{(i)}_{0:k})$. Now for a given $(x^{(i)}_{0:k}, y_{0:k})$,

$p(\theta_k | x^{(i)}_{0:k}, y_{0:k})$ can be considered as the posterior density of the latent param

eter θ_k and can be propagated analytically using a conjugate latent process

as discussed earlier. To see this, note that for a known $x^{(i)}_k$, (26.36a)–(26.36b)

can be transformed into $e^{(i)}_k = e(x^{(i)}_k, y_k) = [g^{\dagger}_k(x^{(i)}_{k-1}, u_{k-1})[x^{(i)}_k - f_k(x^{(i)}_{k-1}, u_{k-1})]d^{-1}_k(x^{(i)}_k, u_k)[y_k - h_k(x^{(i)}_k, u_k)]]^T$,
(26.44) stands for the Moore-Penrose pseudo-inverse of

$g_k(x^{(i)}_{k-1}, u_{k-1})$. Consequently, $p(\theta_k | x^{(i)}_{0:k}, y_{0:k}) = p(\theta_k | e^{(i)}_{0:k})$. So the parameter

posterior follows a conditional conjugate process and the analytical propaga

tion of this posterior can be obtained easily using (26.37)–(26.41).

The joint density (26.43) can be represented as $p(x_{0:k}, \theta_k | y_{0:k}) \approx \frac{1}{n} \sum_{i=1}^n \omega^{(i)}_k p(\theta_k | V^{(i)}_k, \nu^{(i)}_k) \delta(x_{0:k} - x^{(i)}_{0:k})$, (26.45)

where the statistics $\omega^{(i)}_k$, $V^{(i)}_k$, $\nu^{(i)}_k$, and $x^{(i)}_{0:k}$ are evaluated as follows. First,

x_k

(i)

k are sampled from a proposal density $q(x_k | x^{(i)}_{0:k-1}, y_{0:k-1})$. Second, for the

known value $x^{(i)}_k$, $p(\theta_k | x^{(i)}_{0:k}, y_{0:k})$ is updated using the mapping (26.44) to

e_k

(i)

k , and the statistics $V^{(i)}_k$ and $\nu^{(i)}_k$ are updated using (26.39). Finally, using

the recursive relation $p(x_{0:k} | y_{0:k}) \propto p(y_k, x_k | x_{0:k-1}, y_{0:k-1}) p(x_{0:k-1} | y_{0:k-1})$,

weights can be updated as $\omega^{(i)}_k \propto p(y_k, x_k | x_{0:k-1}, y_{0:k-1}) q(x_k | x^{(i)}_{0:k-1}, y_{0:k-1}) \omega^{(i)}_{k-1}$,

where

$p(y_k, x_k | x_{0:k-1}, y_{0:k-1}) = \int p(y_k, x_k | \theta_k, x_{0:k-1}, y_{0:k-1}) p(\theta_k | x_{0:k-1}, y_{0:k-1}) d\theta_k$,

is the marginal predictive distribution of x_k and y_k . This marginal distribution

can be computed by integrating out the unknown parameters, leading to the

predictive distribution of x_k and y_k , and consequently e_k via (26.44). Note

that the predictive distribution $p(e_k | e_{0:k-1})$ is readily available in the form of

(26.40) for the exponential family. The predictor (26.46) can then be obtained

using the transformation of variables in the probability density functions:

$$p(y_k, x_k | x_{0:k-1}, y_{0:k-1}) = |J(x_k, y_k)| p(e_k | v^{(i)}_{k|k-1}, v^{(i)}_{k|k-1}). \quad (26.46)$$

where $J(x_k, y_k)$ is the Jacobian of the transformation (26.44) and $p(e_k | \cdot)$ is

given by (26.40). For further details and an algorithmic summary, we refer to

[32].

Remark 1. If the noise vector e_k follows a multivariate Normal distribu-

tion, with unknown mean μ_k and covariance Σ_k , a Normal-inverse-Wishart

distribution defines a conjugate prior.

Remark 2. This approach is closely related to the works on stationary

parameter estimation using RBPF by [9, 16, 45]. Note that the estima-

tion of stationary parameters is a special case of the above approach with

$\lambda = 1$, which reduces the sufficient statistics update in (26.39) to the form

of [45]. For the stationary parameter, however, this approach is known to

suffer from error accumulations as pointed out by [12]. Here, we note that

for a sequence of $\lambda_k < 1$, $\forall k$, the posterior density $p(\theta_k | x_{1:k}, y_{1:k})$ and thus

$p(y_k, x_k | x_{0:k-1}, y_{0:k-1})$ satisfies the exponential forgetting property that is well

known to mitigate the path degeneracy problem [13]. in Bayesian

26.6 Terrain Navigation Problems

In this section, we outline the problem of positioning an aircraft without the

aid of global positioning systems (GPS). GPS is vulnerable to jamming; we

therefore consider the positioning by using the measurements from alternative

sensors. Specifically, one can position the aircraft using only altitude measure

ments, measurements of the distance down to the ground, measurements of

the aircraft's acceleration, and an elevation map of the visited area.

The navigation problem results in an estimation problem with linear dy

namics, where the aircraft is assumed to follow a constant velocity model, and

a nonlinear measurement equation in the position, $x_{k+1} = [I \ 2 \ T I \ 2 \ 0 \ I \ 2] x_k + [0 \ 2 \ T I \ 2] (w_k + a_k)$ (26.47a) $y_k = h_{map}(x_k) + e_k$, (26.47b)

where T is the sampling period, w_k and e_k are the independent Gaussian

process and the measurement noise, respectively. The first two components

of the state are the position and the next two are the velocity in two dimen

sional Cartesian coordinates (representing north and east). The altitude is

well decoupled from these two directions, and is best handled separately. The

aircraft's body accelerations a_k , are measured and enter into the system as

input. The inertial sensors in an aircraft are generally good enough to pro

vide very accurate information about accelerations. The function h map relates

the terrain altitude to position in the form of a digital terrain database and

is highly nonlinear. The measurements are obtained by deducting the height

measurement of the ground looking (on board) radar from the known altitude

of the aircraft (obtained using an altimeter). Furthermore, the level of the

ground can be expected to be similar in many local regions of the map, hence

the position estimate tends to be a multi-modal distribution, especially after

leaving the relatively uninformative planar areas. This makes the PF a natural

choice when implementing this type of application (as the alternatives such as

extended KF struggles with dealing multimodal distributions). However, ter

rain navigation is often used in systems with limited computational resources

and a straightforward PF implementation is often not feasible.

Fortunately, the model that describes this navigation problem, can eas

ily be identified as an ML/NGSSM; the Rao-Blackwellization conditioned

on the position, x_n , the remaining state-space model describing the veloc

ity, x_l , becomes linear-Gaussian. Applying a RBPF to this problem using

Algorithm 26.1, is then straightforward. The following model components are

identified from (26.11) as $f_n(x_{n:k}) = x_{n:k} F_n(x_{n:k})$
 $= T I^2 h(x_{n:k}) = h \text{ map}(x_{n:k})$ (26.48a) $f_l(x_{n:k}) = 0$
 $2 \times 1 F_l(x_{n:k}) = I^2 H(x_{n:k}) = 0$, (26.48b) only on the
position component of the state, i.e., ,

the nonlinear part of the model. Furthermore, $w_{n:k} \equiv 0$
(i.e., , no process noise

enters directly on the nonlinear part), $w_{l:k} = w_k$, and e_k
is the same as in the

original problem formulation.

A more accurate measurement noise model would also consider
secondary

reflections (e.g., from the tree canopies), resulting in a
bi-modal Gaussian

mixture representation. This could provide further
structure to the problem.

This application has been studied extensively in [5, 31].

26.7 Approximate Rao-Blackwellized Nonlinear Filtering

The implementation of RBPF requires that a subset of the
state vector is

analytically tractable conditioned on the rest of the state
vector. Then one

can marginalize the tractable part to obtain the posterior
distribution of the

non-tractable part using PF. However, for many
applications, this tractability

requirement is very restrictive. For example, consider the
noise adaptive fil

tering where the noise class is not from the exponential
family. For this case,

the unknown noise parameters, in general, do not admit
Bayesian conjugate

priors. Consequently one cannot implement an RBPF for this
case. For such

cases, in recent times, different approximations of the RBPF schemes have

been envisaged in the literature (see e.g., [1, 11, 27, 40, 46]). The basic idea

again is to decompose the whole state space into two (interacting) parts $\mathbf{x}$ and

split the filtering problem into two nested sub-problems.

To illustrate the idea, suppose that the state vector at time k is given by

$(\mathbf{x}_k, \mathbf{z}_k)^T$. The state vector is assumed to follow a Markovian model. Now

given the sequence of observations $\{y_0, y_1, \dots, y_k\}$, we target to evaluate

$p(\mathbf{x}_{0:k}, \mathbf{z}_{0:k} | y_{0:k})$ recursively over k . The target distribution can be decomposed

as $p(\mathbf{x}_{0:k}, \mathbf{z}_{0:k} | y_{0:k}) = p(\mathbf{x}_{0:k} | y_{0:k}) p(\mathbf{z}_{0:k} | \mathbf{x}_{0:k}, y_{0:k})$ (26.49)

We now propose to handle each sub-problem using PFs. First using a single

PF, with N_x particles, we approximate $p(\mathbf{x}_{0:k} | y_{0:k})$. Then we run N_z condi

tional PFs, each with N_z particles, to estimate $p(\mathbf{z}_{0:k} | \mathbf{x}_{0:k}, y_{0:k})$. This is in

essence an approximated RBPF, where we use local Monte Carlo approxima

tions instead of analytical solutions. This idea was proposed in [27, 46] and

also in [40] in the context of noise adaptive PF. A related but slightly different

formulation is considered in [1, 11]. It is important here to emphasize that the

resampling steps for the second sub-problem is local in the state space and

thus can be parallelized. This is a very generic and promising approach for

6 In principle, the state space can be decomposed into more than two parts to implement

an approximate RBPF. in Bayesian inference problems. On one hand, it increases the par

allelism in PF and, on the other hand, it provides flexibility in implementing

specialized PFs with different tailored solutions for individual sub-problems.

26.8 Concluding Remarks

The online (real time) inference problems for general state-space models do not

usually, admit closed form solutions. However, thanks to the modern Monte

Carlo methods, the inference aim can be achieved to an arbitrary accuracy

using particle filtering (PF). Although, PF is a very flexible class of methods

for such inference problems, it is known to suffer from the curse of dimension

ality. For some models, part of the state space admits conditionally tractable

structures. The possibility to exploit such conditional tractability leaves only

a part of the state vector to be targeted using a PF. Consequently, Monte

Carlo based estimation is confined to a space of lower dimension. This in

turn leads to an efficiency improvement of the (filtering) estimator due to the

implications of the Rao-Blackwell theorem. The resulting filtering problem is

popularly known as Rao-Blackwellized particle filtering (RBPF).

In this chapter, we presented a brief review of such RBPF. In particular,

we have outlined a range of existing RBPF models that are popular in practice

and also presented some recent developments in this context. This is written

mainly from the practitioners' point of view and as such, we have not included

any theoretical analysis here. Our coverage is primarily aimed at providing the

model structures admitting such conditional tractabilities. Nonetheless, the

coverage is by no means exhaustive and partially biased towards our interest.

Many other such RBPF may possibly be constructed, e.g., using the exact

filtering sublayer as provided in [36].

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